



ROAD NETWORK TECHNICAL NEEDS ANALYSIS REPORT

UGANDA NATIONAL ROADS AUTHORITY

2019

Executive Summary

The principle objective of this strategic level analysis was to prepare an Investment Plan for the UNRA road network for a five-year period spanning 2019 to 2023. The specific objectives were to:

- Show the impact of the current funding level on the future road network condition, taking into account current committed projects;
- Show the impact of a higher funding scenario on future road conditions;
- Assess the impact of the funding scenarios on maintenance backlog, road user costs, asset value; and
- Prepare a five year optimised Preventive Maintenance and Rehabilitation Plan for the current funding level.

This report contains the results of the analysis and is structured in two components, consisting of a 'situational analysis' and a 'needs analysis'.

Situational analysis

The total length of the national road network investigated in this analysis is based on the length of road with complete visual assessment information in 2018/19. In total 18,232 km of road is investigated of which 3,929 km are paved roads and 14,304 km are unpaved roads. This is 88% of the total road network, being 84% of the paved roads and 89% of the unpaved road network.

Traffic volumes on the national network are very high, approximately 30% of the paved road network carry more than 5,000 vehicles per day and 70% of the unpaved roads carry more than 500 vehicles per day. The high traffic volumes on the unpaved network imply a high need for upgrading to paved standards.

It is evident though, that the impact of motorcycles is high on traffic for all roads. After investigation, there is a significant reduction in traffic volumes when you exclude motorcycles.

For paved roads, traffic above 5000 vehicles/day reduces from 32% to 16% whereas for the unpaved roads, traffic above below 100 Vehicles/day increases from 5% to

29%. It is worthy to note that although the volume of motorcycles is quite significant, we do not entirely incorporate them into the analysis phase. This now calls for an improvement in road safety designs to cater for all cyclist and pedestrians

According to the condition assessment data of 2018, the overall condition of the paved roads is classified as “good”, with 7% in poor and very poor condition, indicating a low rehabilitation backlog. Average VCI is 76%, which is 1% Increase from 2017 and 2% decline from 2015.

The overall condition of the unpaved roads is classified as “fair”. The average condition has largely remained the same since 2017 ranging between 61% and 62%. The poor and very poor roads has increased form 16% in 2017 to 25% and these have high investment needs for upgrade and are functioning with little or no gravel wearing course material.

Needs analysis

This analysis determined the impact of the current funding levels considering the following intervention treatments: reseal, rehabilitation, regravelling and upgrading of unpaved roads to paved standards.

The analysis derived at the following outcomes:

- The current funding level of US\$ 1,054 per annum is sufficient to maintain the road network, at current conditions, and will increase the performance levels over the analysis period minimally.
 - The optimal allocation of the current funding to paved roads is 10% (US\$ 102 million per annum); and these roads will be maintained in the current condition category of “good” over the next 5-10 years.
 - The remaining 90% (US\$ 953 million per annum) of the current funding will increase the gravel thickness of unpaved roads to over 100mm over the 5-10 year analysis period. US\$ 25 million per annum should be allocated to regravelling, and US\$ 928 million per annum towards upgrade to paved standards.
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Whereas the network can successfully be maintained with the current funding, the excessive need for upgrading of unpaved roads to paved standards is not fully met. Therefore, when comparing the analysis allocation between the current received and a higher funding scenario ('required funding' of US\$1,243 million per annum), the main difference in the optimal funding allocation is the increased funding allocation toward unpaved roads, and specifically toward upgrading to paved standards.

All unpaved roads triggered for upgrade shall require more funding to keep them in a motorable state as unpaved roads due to the increased frequency of maintenance.

It is therefore critical that additional funding be sourced to upgrade them, as they can no longer be maintained as unpaved roads in relation to overrun costs of maintenance.

Furthermore there is need to implement preventive maintenance programs to preserve the asset value and life of the paved roads.

Lastly, It is proposed that a low cost delivery mechanism to pave roads be adopted especially for roads with less than 1000 vehicles per day.

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1 Introduction

This document is structured in two parts. The Situational Analysis; which gives an understanding of the composition of the road network, traffic information as well as the current condition; and the Needs analysis; which depends on the findings from the situational analysis as inputs and then outputs preventive maintenance, rehabilitation and upgrading needs of the National road network. This is also referred to as an Investment Plan or National Road Investment Plan (NRIP).

The document further summarises the life cycle cost analysis (LCCA) conducted for the National road network for a ten year analysis period. The purpose of the LCCA was to prepare a five-year National Roads Investment plan for the network. The analysis was undertaken using the dTIMS (Deighton Total Infrastructure Management System) which was customised with the World Bank's Highway Development and Management (HDM-4) pavement performance prediction algorithms. The optimisation function in (dTIMS) predicts future pavement performance and consequences of alternative funding and policy scenarios. Multi-year preventive maintenance, rehabilitation and upgrading programs were then generated for given funding scenarios. These programs are used as basis for compiling the annual works programs.

The investment needs described in this analysis and report are relevant only for preventive maintenance, rehabilitation and upgrading to paved standards. Needs for interventions such as routine maintenance, capacity and safety improvements, structures, etc are not included in the funding needs determined by this analysis.

2 Report Outline

The report is structured in two major components, each consisting of a number of chapters. The components are:

- Situational analysis (chapters 3 to 6)
- Needs analysis (chapters 7 to 13)

Table 2-1 provides the report outline with the contents per chapter.

TABLE 2-1: REPORT OUTLINE	
CHAPTER	CONTENT
Error! Reference source not found.	The road network under UNRA's jurisdiction is described in terms of the pavement type as well as the roads for which data is available that will be used for the analysis.
4	The extent of traffic usage of the road network is presented.
Error! Reference source not found.	The current condition of the paved road network, based on visual assessment data, is presented and compared with historic data. Measured roughness data is also included.
6	The current condition of the unpaved road network, based on visual assessment data, is presented and compared with historic data.
3	Input parameters and background to the life cycle costs analysis (LCCA) are discussed (treatments, treatment unit costs, budgets, performance prediction models, etc).
8 - 10	These chapters investigate the Technical Needs scenario to determine the funding level required to remove the current backlog in maintenance and rehabilitation needs. Other funding scenarios are investigated in terms of the optimisation of funds towards maintenance and rehabilitation of paved and unpaved roads.
11	The different funding scenarios are compared in terms of paved road condition, gravel thickness and the differences in fund allocations are assessed.
12 -13	Vehicle operating costs and excess user costs for different funding scenarios are assessed.

14	The current asset value, change in asset value and maximum asset value are investigated.
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3 The UNRA Road Network

Summary:

UNRA now has 20,856km of roads under its jurisdiction.

However, at the time of analysis, and for purposes of this report

UNRA had 20,794 km of road under its jurisdiction (March 2019).

23% (4,683 km) paved, and 77% (16,111 km) unpaved.

3,929 km (84%) of paved roads was analysed in this report.

14,298 km (89%) unpaved roads was analysed.

At the time of analysis and according to the UNRA RMS (Road Management System), the total length of the national road network is 20,794 km. Of these 4,683 km are paved roads and 16,111km are unpaved roads. Furthermore, according to functional classification, 13% of this network is Class A, 14% Class B and 73% is Class C as indicated below.

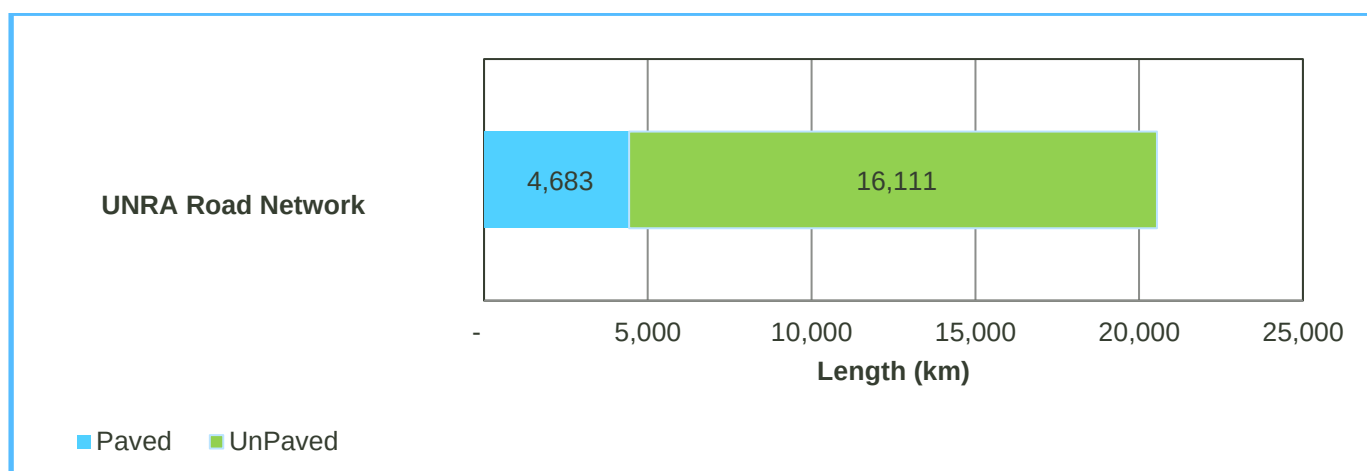


Figure 3.1: Split between Paved (23%) and Unpaved (77%) Road lengths of UNRA Network (2018)

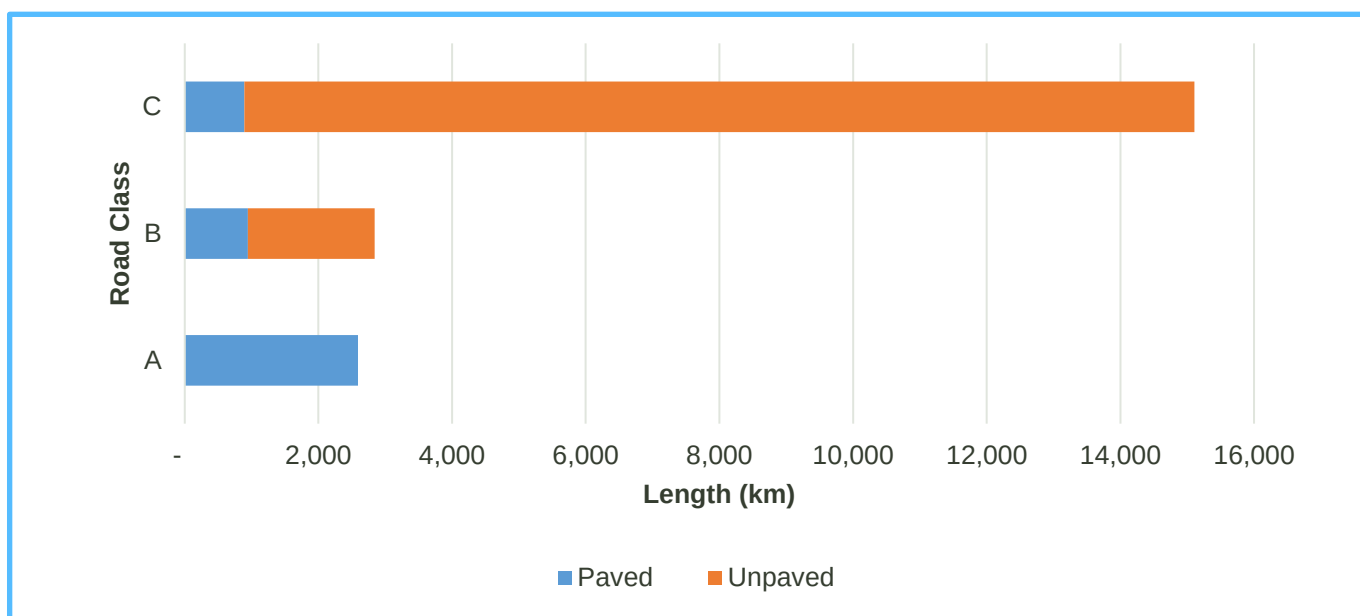
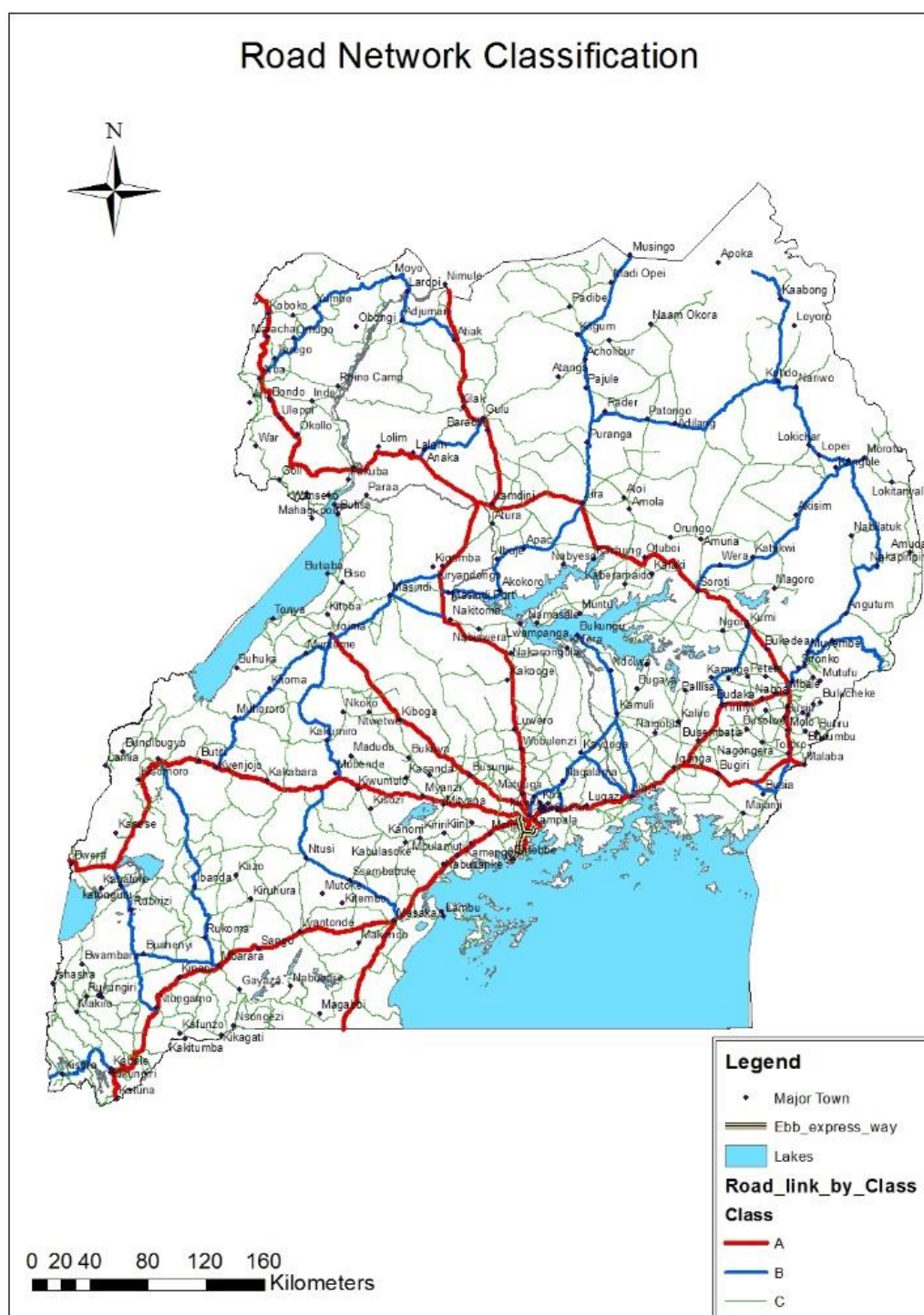


Figure 3.2: Functional Classification of UNRA Road Network.

Updated Network Functional Classification (June 2019)

Functional Classification	Paved	Unpaved	Length (Km)	%	Description
A	2594	0	2594	12	International Trunk Roads
B	1148	1695	2843	14	National Trunk Roads
C	1187	14147	15334	74	Primary Roads
M	85	0	85	0.4	Expressways
Total			20856		



The total length of the national road network investigated in this analysis is based on the length of road with complete visual assessment information in 2018/2019. In total 18,232 km of road is investigated of which 3,929 km are paved roads and 14,304 km are unpaved roads. This is 88% of the total road network, being 84% of the paved roads and 89% of the unpaved road network.

Note that although it is preferable that data for the full road network be available for any analysis, this is influenced by roads being under construction at the time of the surveys, and changes or corrections to the roads register.

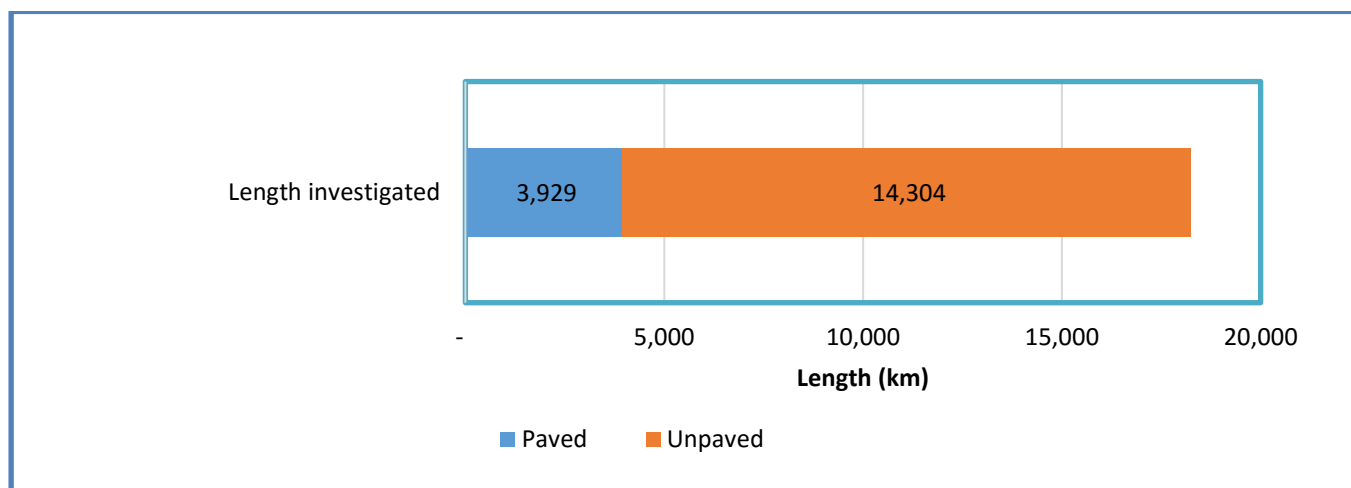


Figure 3.3: Paved and Unpaved road lengths investigated (km), 2018

4 Traffic Information

Summary:

- ***14,303 million vehicle-kilometres are travelled annually on the UNRA road network.***
- ***7,742 million vehicle-kilometres are travelled annually on paved roads and 6,561 million vehicle-kilometres annually on unpaved roads.***
- ***30% of the paved network carries more than 5,000 vehicles per day and 70% of unpaved roads carries more than 500 vehicles per day.***

The traffic data was collected by UNRA staff and Data Collection Consultants on framework contracts in the second and third quarter of FY 2018/2019 at 310 No MCC (Manual Classified Counting stations). This data was used in the TIS (Traffic Information System) to allocate traffic counts to every road link on the network based on the mother / daughter approach followed in the TIS. This follows a rigorous clustering approach to associate various links to others where traffic was counted.

Traffic information is then expressed as the Annual Average Daily Traffic (AADT), travelling on each road link of the road network. Furthermore, the product of the AADT on a road link and its length represents the daily vehicle-kilometres travelled on that link. Summarised for the road network, they represent the total distance travelled by vehicles on the road network on a daily or yearly basis. Approximately, 14,303 million vehicle-kilometres are travelled each year on the national road network (*12,977 Million Veh-km in 2017*). Approximately 7,742 million vehicle-kilometres are travelled annually on paved roads (*7,418 Million Veh-km 2017*) and approximately 6,561 million vehicle-kilometres annually on unpaved roads (*5,579 Million Veh-Km 2017*).

Figure 4-1 shows that although the investigated paved road network slightly above 20% of the total road network carries over 50% of the network's traffic. A focus on the maintenance and rehabilitation of paved roads will therefore benefit more road users per kilometre than unpaved roads. This will guarantee savings in excess user costs, vehicle-operating costs and reduce travel times.

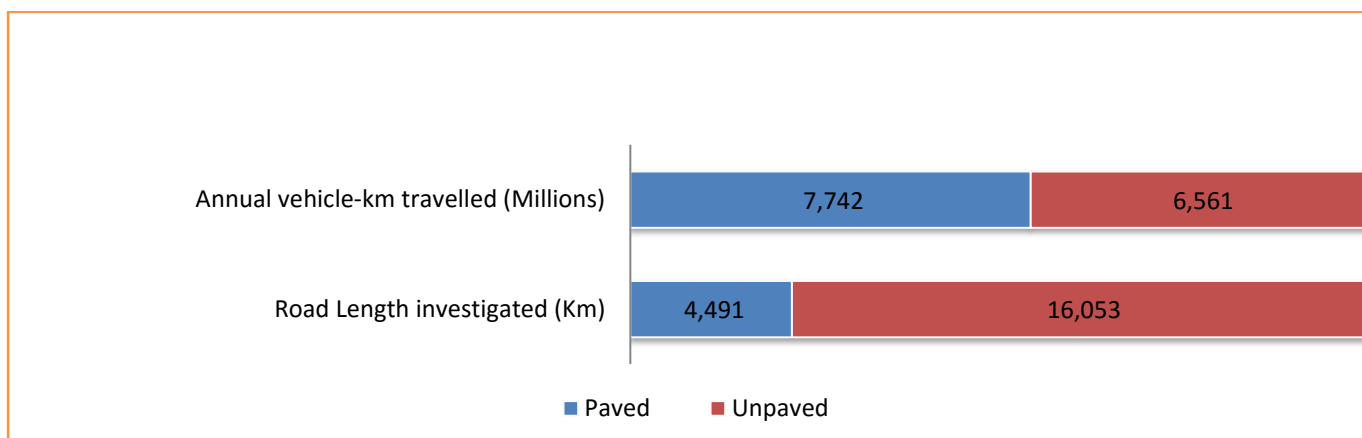


Figure 4-1: Network Length investigated Vs vehicle-kilometres travelled, 2018

4.1 Traffic per road type

The distribution of traffic volumes for paved and unpaved roads is given in Figure 4-2 and Figure 4-3 respectively. These figures show the portions of network carrying low to high categories of traffic. 70% of the paved network carries less than 5,000 vehicles per day while 21% carries 5,000 to 10,000 vehicles per day and the other 11% carries more than 10,000 vehicles per day. For unpaved roads, approximately 40% carries between 1,000 and 5,000 vehicles per day and 1% carries more than 5,000 vehicles per day. Overall, 70% of unpaved roads carries more than 500 vehicles per day. An increment of 5% from 2017.

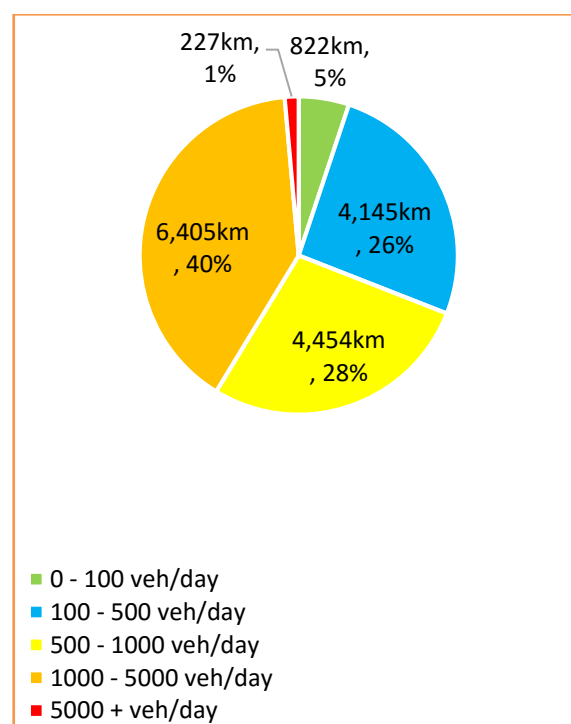
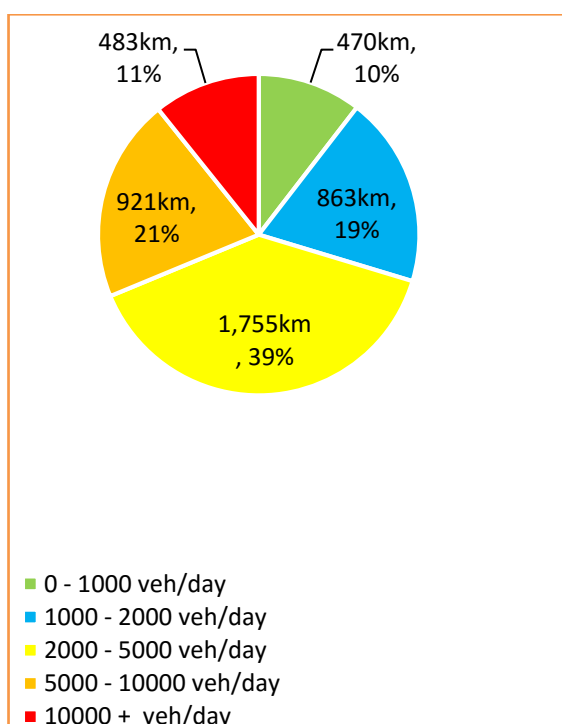
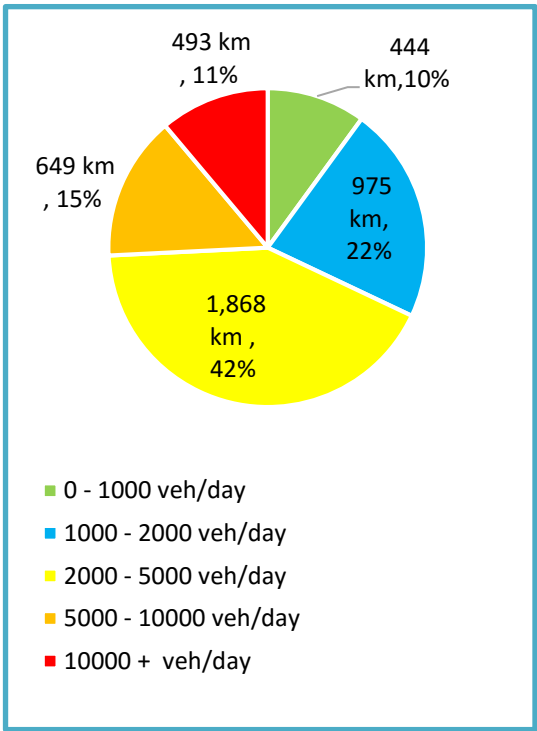


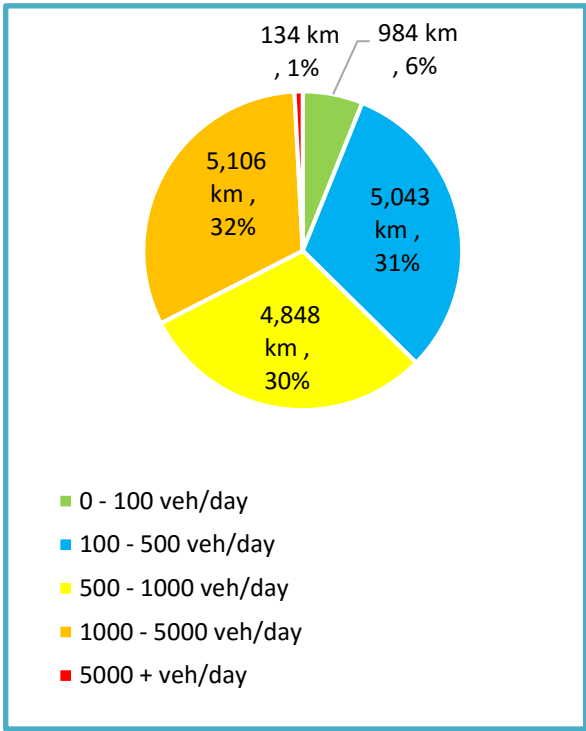
Figure 4-2: Traffic on paved roads, 2018

Figure 4-3: Traffic on unpaved roads, 2018

2017 TRAFFIC



2017 Traffic on paved roads



2017 Traffic on Unpaved roads

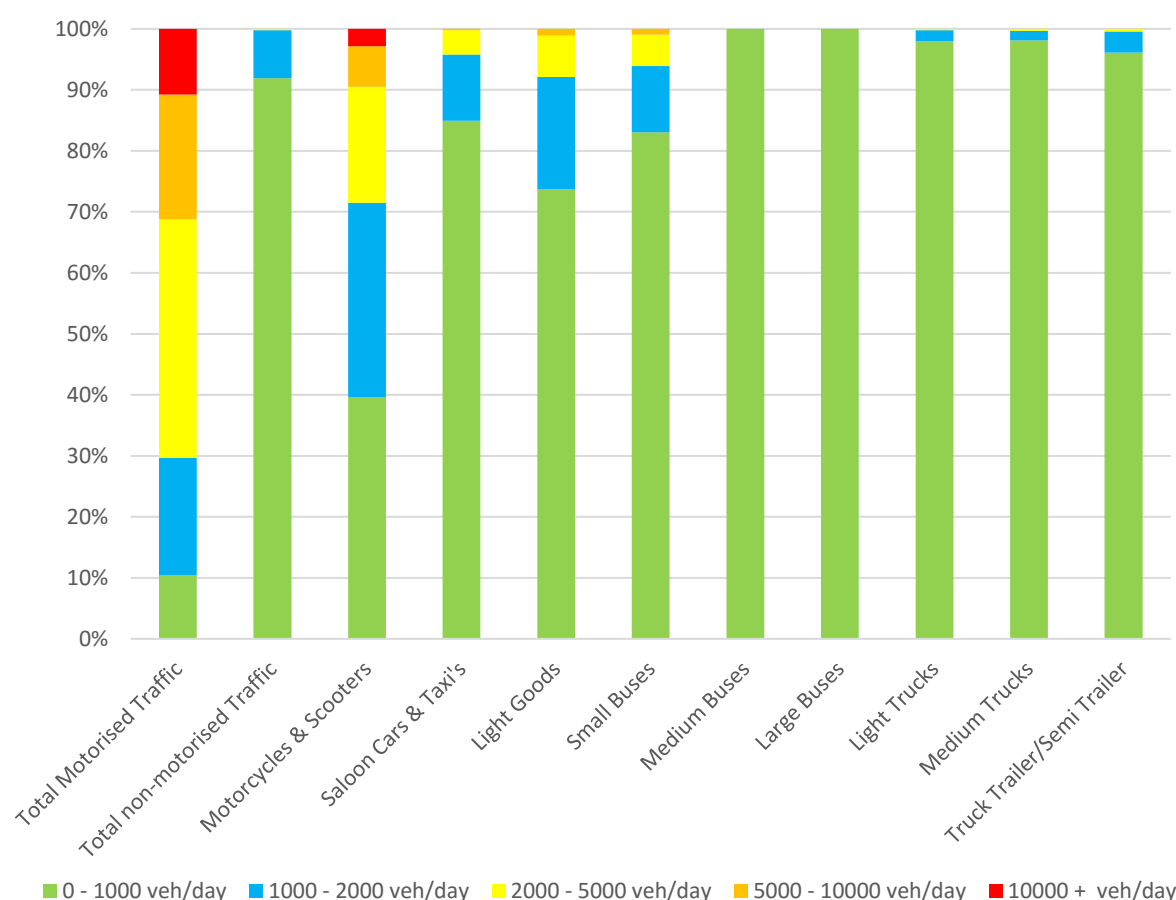


Figure 4-4: Traffic distribution on paved roads per Vehicle Type, 2018

Figure 4-4 confirms that 30% of the traffic is composed of volumes more than 5,000 vehicles per day (refer Total Motorised Traffic)(5% more than 2017). The figure also shows a breakdown of the vehicle types in the traffic stream, confirming the extensive amount of high volume travel via motorcycles and scooters (Boda-Bodas), followed by light goods vehicles and small buses(Minibuses and Matatus). In general, travel exceeding 1,000 vehicles per day is limited to Saloon cars &Taxis, Light goods, small buses, Light/Medium trucks and Truck Trailer/Semi Trailer vehicle types, while travel less than 1,000 vehicles per day is limited to Medium and Large Buses.

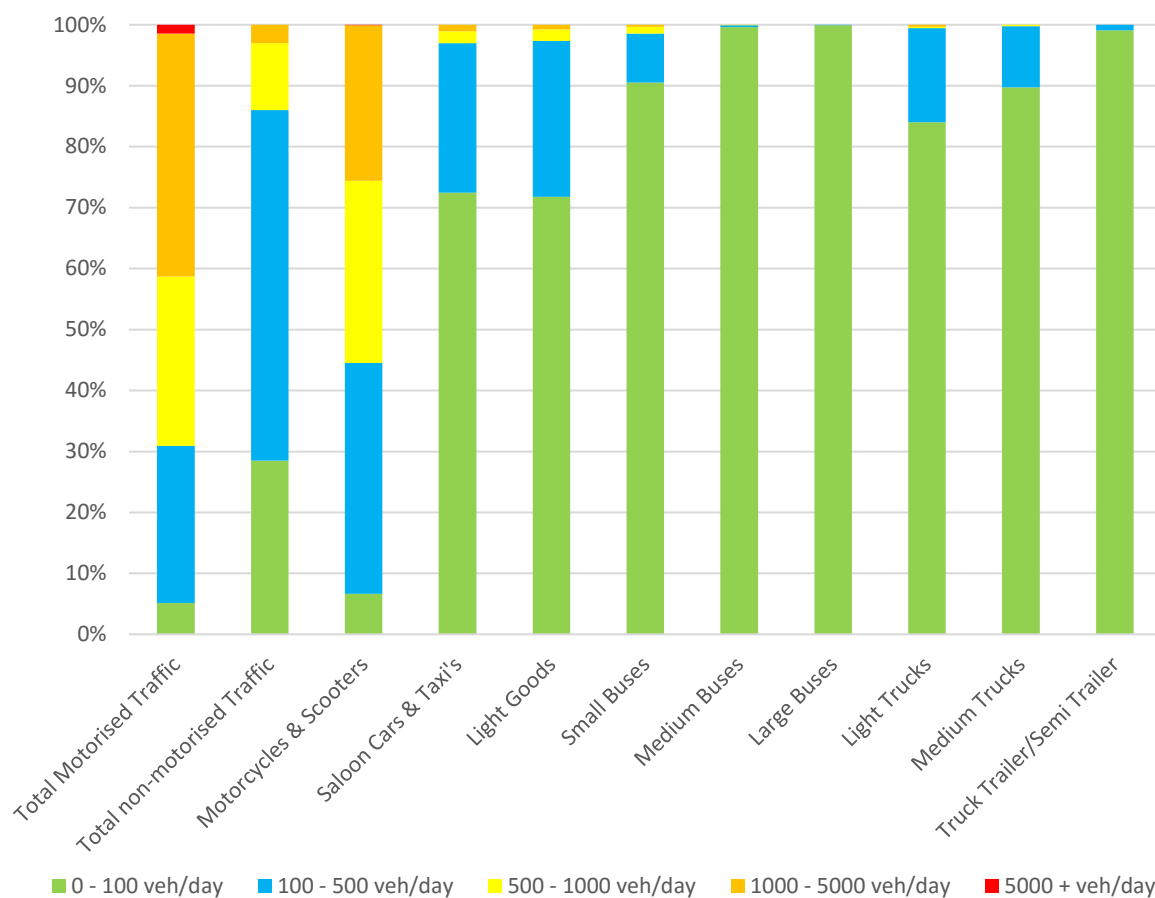
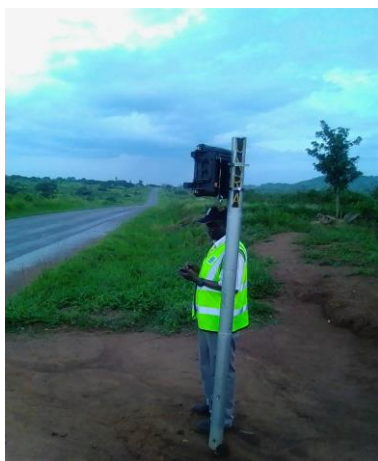


Figure 4-5: Traffic distribution on unpaved roads per Vehicle Type, 2018

For unpaved roads, Figure 4-5 confirms that 40% of unpaved road travel concerns volumes between 1,000 and 5,000 vehicles per day. Similar to paved roads, most of the travel is by motorcycles and scooters (Boda-Bodas) followed by Light Goods and Saloon Cars & Taxi's. With an exception of Motorcycles and Scooters, 70% of travel for all vehicle classes are below 100 vehicles per day. See Appendix A (Traffic 2018).

Traffic Data Collection



Traffic excluding Motorcycles

It is evident that the impact of motorcycles is high on all roads. As seen from the Table 4.6 below, there is significant reduction in traffic volumes when you exclude motorcycles.

For paved roads, traffic above 5000 vehicles/day reduces from 32% to 16% whereas for the unpaved roads, traffic above below 100 Vehicles/day increases from 5% to 29%. It is worthy to note that although the volume of motorcycles is quite significant, we do not entirely incorporate them into the analysis phase.

Paved Roads			Unpaved Roads		
	AADT Incl Motorcycles	AADT Excl Motorcycles		AADT Incl Motorcycles	AADT Excl Motorcycles
0 - 1000 veh/day	10	30	0 - 100 veh/day	5	29
1000-2000 veh/day	19	30	100 - 500 veh/day	26	54
2000-5000 veh/day	39	25	500 - 1000 veh/day	28	9
5000-10000 veh/day	21	12	1000 - 5000 veh/day	40	7
10000+ veh/day	11	4	5000 + veh/day	1	0

TABLE 4-6: **Traffic Comparison**

5 Paved Roads Network Condition

Summary:

- *The average paved road VCI based on the 2018 visual assessment data is 76%.*
- *Moderate to severe 'All Cracking' is noted on 10% of the paved, a 15% reduction from 2017.*
- *15% of roads have severe to moderate ravelling (surface disintegration) and immediate maintenance is required on these roads. This has doubled from 2017*
- *There is severe to warning roughness on 15% of the road network, a 5% increase from 2017.*
- *12% of the road network exhibits rutting in moderate to severe state, a 13% reduction from 2017.*
- *Overall, the paved road network is in good condition with less than 10% in poor and very poor condition.*

5.1 Background to the VCI

The assessment data, expressing the carriageway condition through the assessment ratings of pavement defects are utilised to calculate a single Visual Condition Index (VCI). This VCI is calculated for each paved road segment (typically of 1 km length) that was surveyed. VCI is a percentage index ranging between 0 and 100; 0 represents a road segment in very poor condition and 100 represents a road segment in very good condition. During this analysis the VCI is used as follows:

- An average network VCI is calculated representing the average condition of the paved road network or any sub-network (e.g. regional network). This average network VCI is weighted by length and is used to describe/investigate the average condition of the road network/sub-network.
- The VCI is furthermore used to categorise the condition of a paved road segment, the paved road network or sub-network into one of five condition categories. The condition categories adopted are:
 - Very Good = 86% to 100%
 - Good = 71% to 85%

- Fair = 51% to 70%
- Poor = 31% to 50%
- Very Poor = 0% to 30%

5.2 Visual Assessment and Distress Graphs for Paved Road network

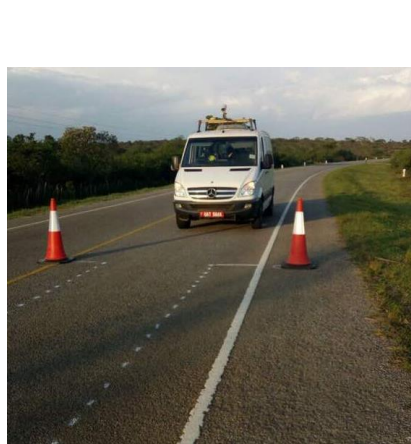
The condition of flexible pavements was assessed by visual inspection and by profile measurement equipment. Paved visual assessments and paved roughness measurements were executed in January/February 2019 using in-house capacity and machinery. The Pavement Management System (PMS) of the UNRA RMS was updated with this data following data validation.

Figure 5-1 is a schematic representation of the distress items showing the total percentage of paved roads rated as

- 1  (very good or non-existent),
- 2  (slight, good or isolated occurrence),
- 3  (discernible, fair or moderate occurrence),
- 4  (warning, poor or general occurrence) and
- 5  (severe, very poor or extensive occurrence).

These ratings can indirectly be related to the proportion of the paved road network with  none,  isolated,  moderate,  warning and  severe problems.

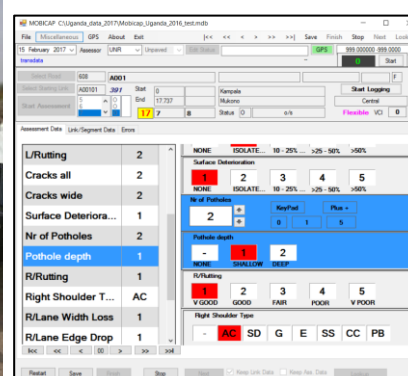
Condition Assessment



ROMDAS Van



Mobicap Tool for assessment



Assessment Data	Distress Rating
L/R Rutting	2
Cracks all	2
Cracks wide	2
Surface Deterioration	1
Nr of Potholes	2
Pothole depth	1
R/Rutting	1
Right Shoulder T...	AC
R/Lane Width Loss	1
R/Lane Edge Drop	1

According to international standards, a “healthy” road network has less than 10% poor and very poor roads. Attention should thus be given to the distresses and condition

ratings exceeding this benchmark for warning and severe problems. According to the visual assessments and automated measurements the following conclusions can be made with regards to the paved roads of UNRA:

Cracking	■ Moderate to severe 'All Cracking' is noted on 10% of the paved roads with another 25% exhibiting isolated cracking. There are moderate to severe ratings for wide cracks on 15% of the paved road network. Wide cracking is a particular concern because these cracks indicate the disintegration of the structural layers in the wheel paths. Wide cracks also initiate the formation of rutting. Preventive measures should be considered as soon as possible to delay further deterioration with the development of secondary distresses.
Ravelling	■ Ravelling is the loss of aggregate material from the road surfacing. In seals, this manifests through stone loss, in asphalts through disintegration of the surfacing. ■ 15% of roads have severe to warning ravelling and immediate maintenance is required on these roads. Future maintenance of roads with moderate ravelling is crucial since it accounts for 15% of roads which could easily deteriorate to warning condition.
Potholes	■ 2% of roads are in a moderate to severe condition due to the occurrence of potholes. Isolated potholes occur on a further 22% of the network.
Roughness (measured)	■ There is mostly warning roughness on more than 15% of the road network. These roads typically require some form of rehabilitation. ■ A substantial portion of the network (roughly 50%) is in a moderate condition regarding roughness. Although some of these roads might not require immediate attention it is crucial to monitor the condition continuously and apply the right treatment at the right time to delay further deterioration.

Rutting	12% of the road network exhibits rutting in moderate to severe state and 75% with isolated rutting.
Width Loss	There is minimal width loss (due to edge break) on the network whether shoulders exist or not.
Edge Drops	As for width loss, edge drops are severe to moderate where paved shoulders exist, on 30% of the network. This indicates to edge drops more than 100mm height.
Bleeding	
Shoulder condition	Less than 5% of the paved network exhibits bleeding in warning and moderate states with 20% having isolated bleeding.
Drainage on road	The 2m width is available to the road user as emergency stopping area. There are isolated problems on 25% of the road network and less than 5% in moderate to severe condition.
Drainage off road	- This refers to surface drainage and ability of road to drain water from the surfacing. Generally not a problem, only about 10% in moderate condition. - Referring to condition of side drainage, mitre drains and cross drainage underneath the road. 10% in a moderate to severe conditions.

Paved Roads - 2018

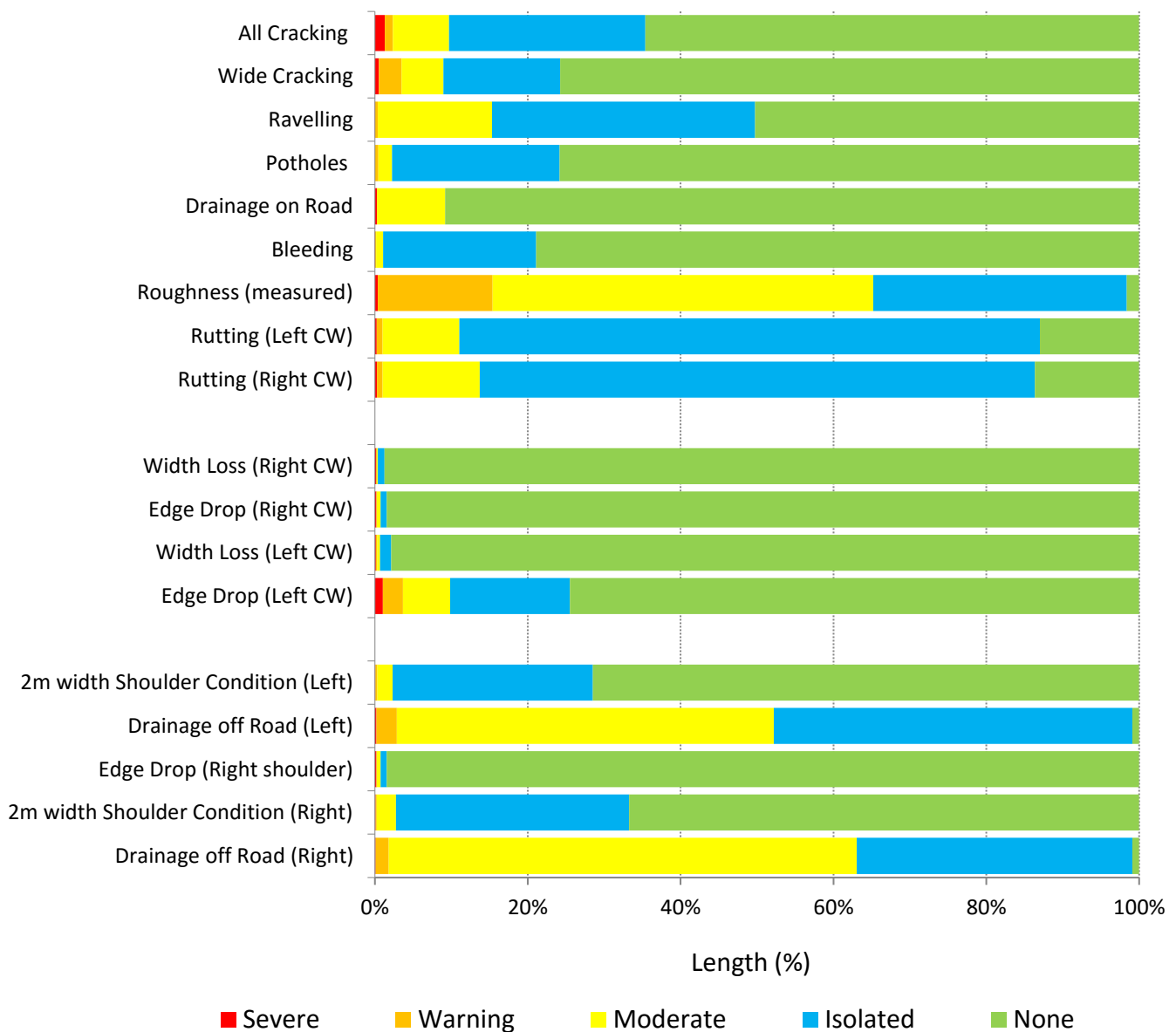
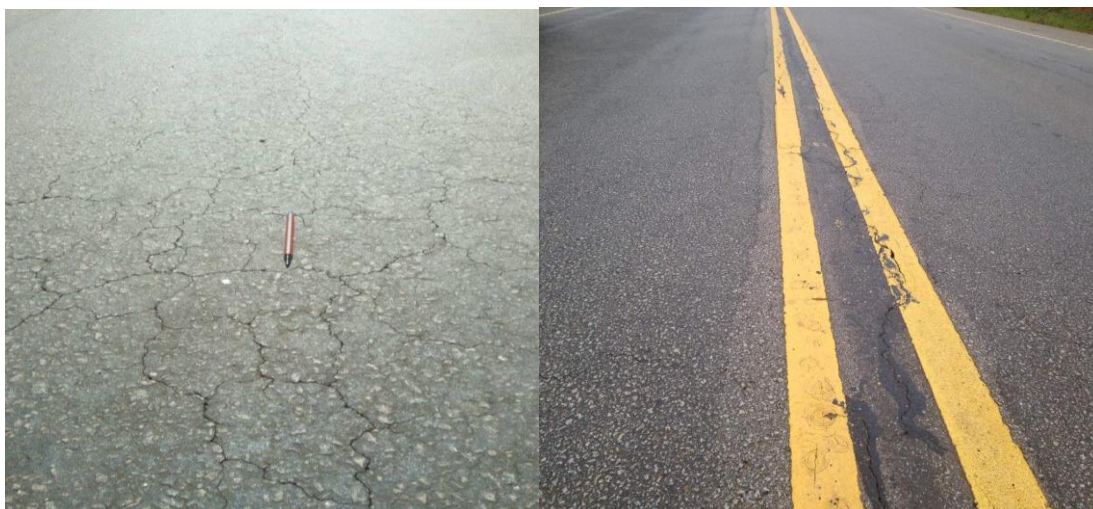


Figure 5-1: Measured values and distress ratings of paved roads, 2018

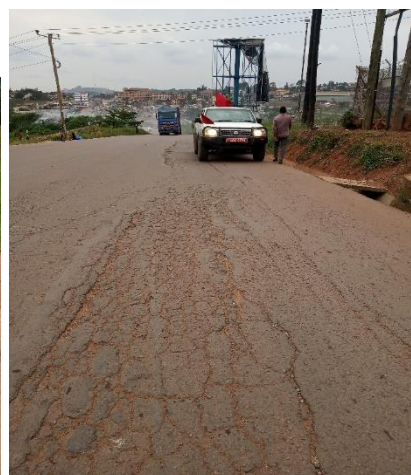
In summary, the paved network is in good condition. From a pavement management perspective, roads appear to be structurally sound, but cracking, roughness and rutting indicate some structural deterioration on less than 5% of the network (rated as severe to warning). A preventive maintenance regime of timely patching, crack sealing, diluted emulsions and resurfacings will contribute significantly to delay the structural deterioration by preventing water from entering the pavement structures.

Below are pictures showing the deterioration on paved roads.



■ Busega – Masaka

Mbarara - Kabale



■ Masaka – Kyotera Km 14
24+300

Nambole Access

Kyotera - Mutukula Km



■ Busega - Bujuuko
44+500

Kiganda - Kitenga 111+186

Buwenge – Kamuli Km



■ Namutere - Busia



Masaka – Lyantonde Km 189+800

5.3 Average VCI for Paved roads

The average paved road VCI based on the 2018/2019 visual assessment data is 76%, which places the road network in the “good” condition category. This is a one percentage point increase on the 2017 average VCI.

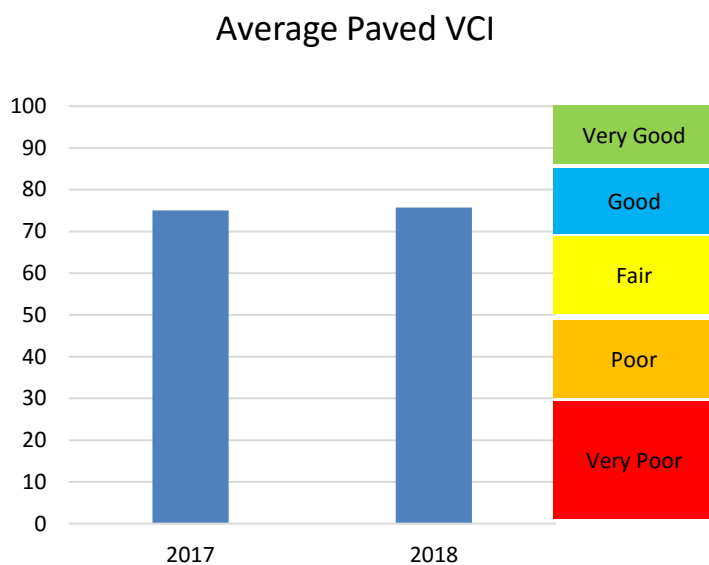
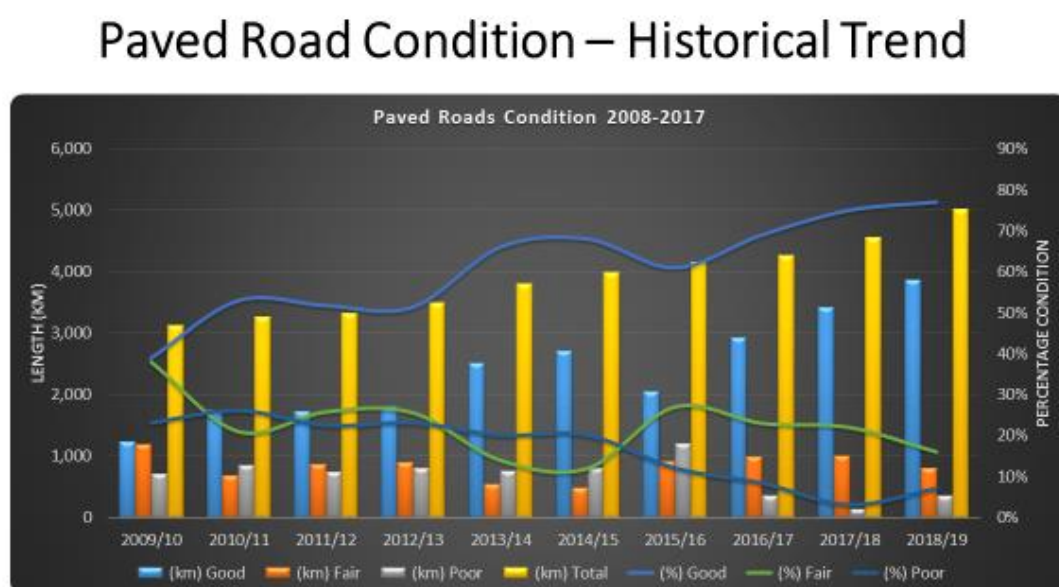


Figure 5-2: Average Paved VCI comparison



5.4 Condition distribution graphs for VCI of Paved Roads

The figures below show that the percentage of roads in very poor and poor condition has increased from to 3% in 2017 to 7% in 2018. The roads in very good condition have increased by 8% since the assessment in 2017. The roads in fair and good condition has decreased by 6%.

Overall, the paved road network is in good condition with less than 10% in poor and very poor condition, indicating a low backlog for rehabilitation. However, the 16% roads in fair condition indicate significant backlog for preventive maintenance.

If this is not addressed in time, these roads will be subject to Moisture Accelerated Damage and will convert to very poor condition in a short time thus increasing the rehabilitation backlog in a very short time. As such, we need to start undertaking preventive maintenance of paved roads to reduce overall costs of rehabilitation, which are 10 times more than the former.

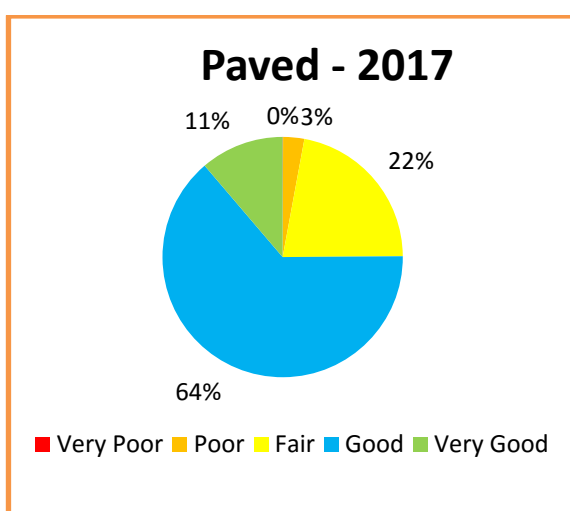


Figure 5- 3: Paved VCI for 2017

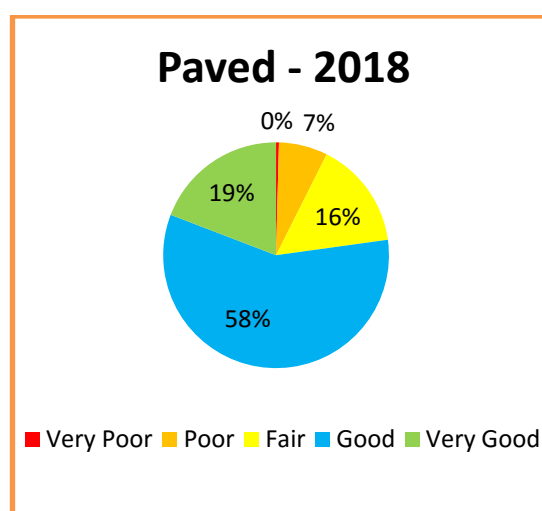


Figure 5- 4: Paved VCI for 2018

6 Unpaved Road Network Condition

Summary:

- *The average unpaved road VCI based on the 2018 data is 61% which categorises the road network in the “fair” condition category.*
- *The gravel thickness ratings indicate the need for urgent re-gravelling on 25%. 4% has no gravel at all and 21% has gravel less than 50mm.*
- *The roughness ratings on the unpaved network indicate severe to moderate condition states on 36% of the road network.*
- *Roads with rutting in a moderate to severe condition accounts for almost 50% of the network.*
- *37% of all roads have severe to moderate corrugations (20% - 2017).*
- *Erosion is very common with almost 52% of roads showing the occurrence of severe to moderate erosion.*
- *Drainage condition is poor and very poor on 17% of the network (8% - 2017).*
- *% of Roads in very poor and poor condition has increased from 15% in 2017 to 25% in 2018.*

6.1 Background to Unpaved VCI

The assessment ratings are utilised to calculate a Visual Condition Index (VCI) similar to the VCI of paved roads. This unpaved VCI is calculated for each unpaved road segment that was surveyed. VCI is a percentage index ranging between 0 and 100; 0 represents a road segment in very poor condition and 100 represents a road segment in very good condition. During this analysis the VCI is used as follows:

- An average unpaved network VCI is calculated representing the average condition of the unpaved road network or any sub-network. This average network VCI is weighted by length and is used to describe/investigate the average condition of the road network/sub-network.
- The VCI is furthermore used to categorise the condition of an unpaved road segment, the unpaved road network or sub-network into one of five condition categories. The condition categories adopted are:

- Very Good = 86% to 100%
- Good = 71% to 85%
- Fair = 51% to 70%
- Poor = 31% to 50%
- Very Poor = 0% to 30%

6.2 Visual assessment of Unpaved Road network

Unpaved visual assessment data was collected in 2018/2019 and the PMS updated with these data.

The assessment data expresses the condition and characteristics of the wearing course material and the functional condition of the gravel roads as a set of rating scores, rating these as;

- 1  (very good or non-existent),
- 2  (slight, good or isolated occurrence),
- 3  (discernible, fair or moderate occurrence),
- 4  (warning, poor or general occurrence) and
- 5  (severe, very poor or extensive occurrence).

These ratings can indirectly be related to the proportion of the unpaved road network with  none,  isolated,  moderate,  warning and  severe problems.

According to the visual assessments, the following conclusions can be made about the unpaved roads (refer Figure 6-1):

Gravel thickness	■ The gravel thickness ratings indicate the need for re-gravelling. This need is currently high, with 4% having no gravel, another 21% having less than 50mm gravel wearing course and severe exposure of the sub grade. A further 51% is in the moderate condition state, with subgrade exposed in local areas (average of 50 to 100mm gravel thickness).
Material quality	■ 13% of roads was rated with warning material quality (poor particle size distribution). A further 57% exhibits cracking, loose material or stones clearly visible, thus rated as material quality of moderate state.

Roughness	■ The roughness ratings on the unpaved network indicate 36% of the road network to be warning to moderate condition. This will result in high road user costs.
Potholes	■ The high occurrence of potholes especially in warning to moderate states (31%) and over 40% in isolated state is indicative of inadequate maintenance.
Rutting	■ Roads with rutting in a moderate to warning condition accounts closely to 50% of the network. These roads are in need of blading and re-graveling.
Corrugations	■ 36% of all roads have warning to moderate corrugations. A further 47% has isolated corrugations. Although these roads are in need of blading, there seems to be a need to investigate whether material properties contribute to the formation of corrugations.
Erosion	■ Erosion was rated serious on over 4% of road network. There is a need to repair erosion more regularly.
Drain condition and formation level	■ Drainage condition and formation level indicate the ability of roads to drain water away, transversely underneath and longitudinally next to them. Roads with severe or warning formation levels, would act as a canal or be level with ground level, this is the case for almost 2% of the unpaved road network. Furthermore, although drainage condition is severe close to 6% of the network, moderate drainage conditions exist on approximately 50% of the network.

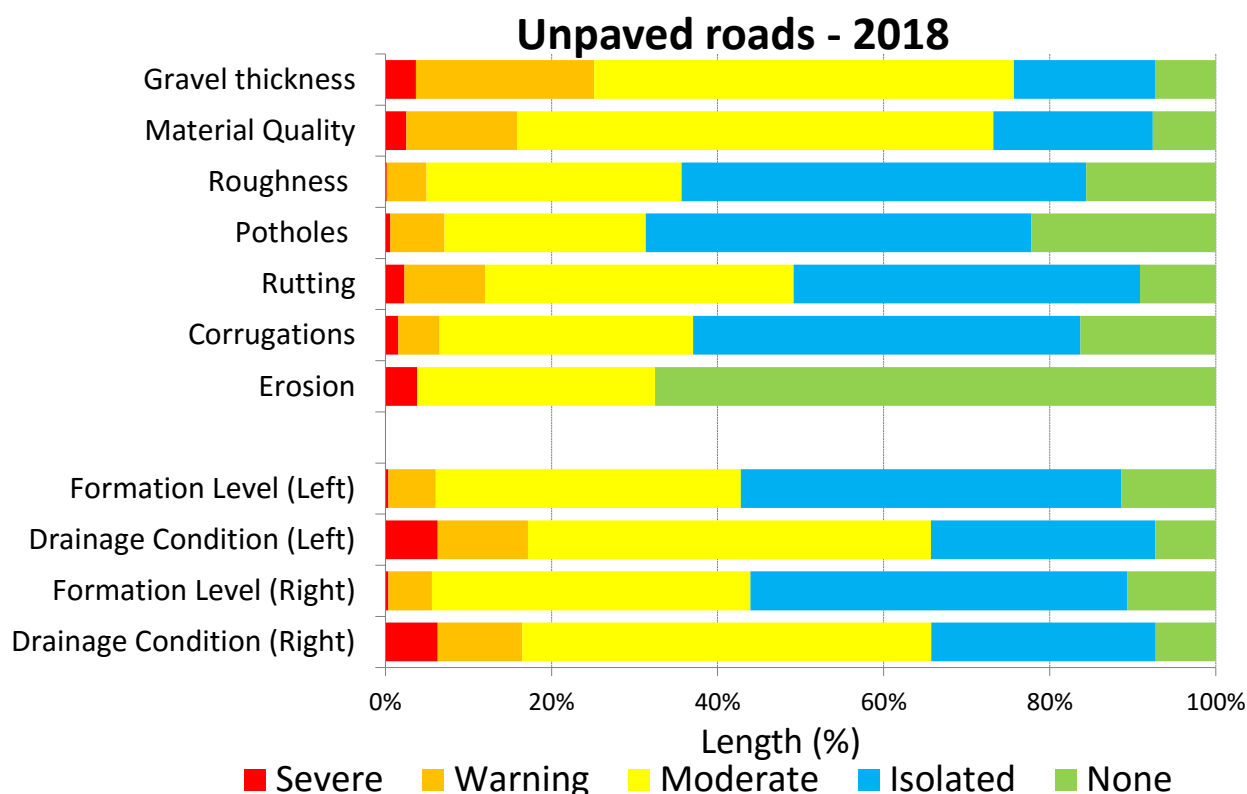


Figure 6-1: Distress, characteristic and condition ratings of unpaved roads, 2018

In summary, the unpaved network is in fair condition. From a road user's perspective, material quality and possibly non-sufficient maintenance regime contribute to the development of rough roads with potholes, rutting and corrugations. From an unpaved roads management perspective, gravel thickness problems, low formation levels and poor drainage conditions result in erosion, wetting of the gravel layers and subgrade, which result in potholes, and rutting. A well-planned regravelling programme should be supported by an effective maintenance regime of blading (to maintain cambers and limit roughness), cleaning of side drains, mitre drains and cross drainage.

6.3 Average VCI for Unpaved roads

The average unpaved road VCI based on the 2018 data is 61%, which categorises the road network in the "fair" condition category; this is a 1% decline as compared to the average VCI in 2017.

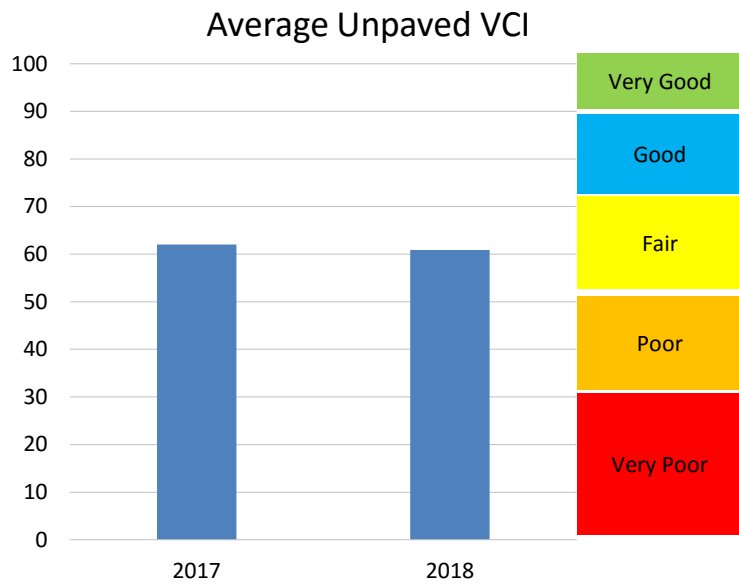
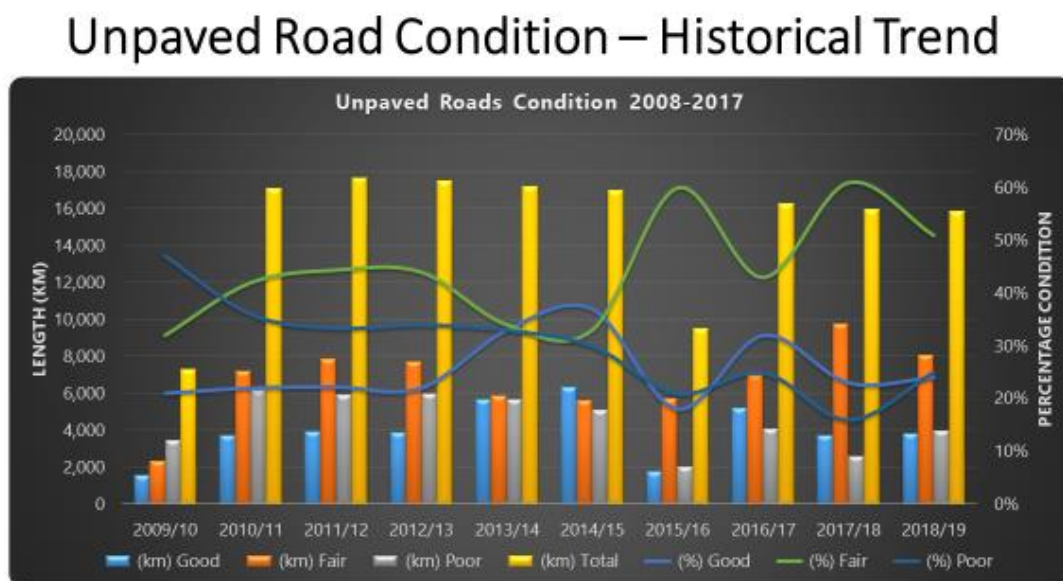


Figure 6-2: Average Unpaved VCI comparison



6.4 Condition distribution graphs for the VCI of Unpaved roads

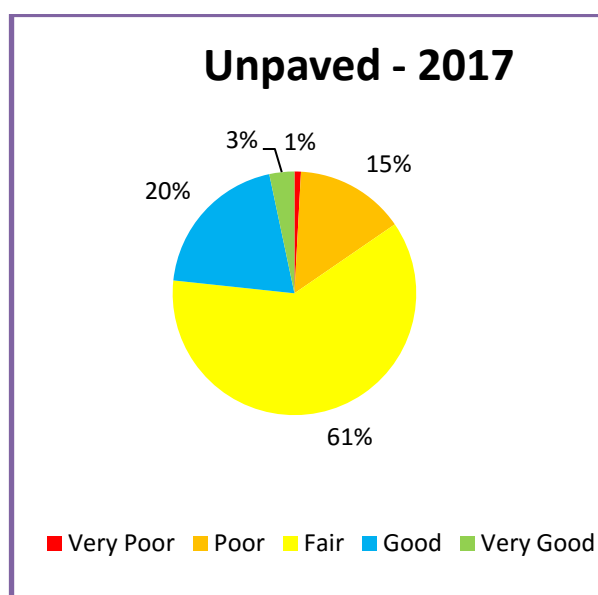


Figure 6-3: Unpaved VCI distribution for 2017

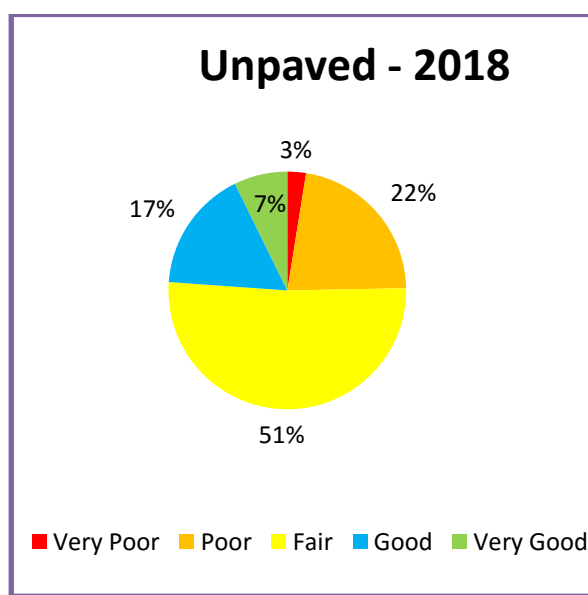


Figure 6- 4: Unpaved VCI distribution for 2018

The figures show that the percentage of roads in very poor and poor condition, which decreased from 25% in 2015, has increased from 16% in 2017 back to 25% in 2018. The roads in fair condition, which increased from 43% in 2015, has decreased from 61% in 2017 to 51% in 2018. The condition of roads in good and very good condition has slightly increased by 1% from previous assessments.

In summary, the unpaved network is in fair condition. From a road user's perspective, material quality and possibly non-sufficient maintenance regime contribute to the development of rough roads with potholes, rutting and corrugations. From an unpaved roads management perspective, gravel thickness problems, low formation levels and poor drainage conditions result in erosion, wetting of the gravel layers and subgrade, which result in potholes, and rutting. A well-planned gravelling programme should be supported by an effective maintenance regime of blading (to maintain cambers and limit roughness), cleaning of side drains, mitre drains and cross drainage.

The situational analysis now sets the scene for the roads needs analysis.

7 The Needs Analysis

Background:

HDM-4 pavement performance prediction models are used to model the future performance of the UNRA roads.

The objective of the optimisation function of this LCCA is to preserve the network condition.

Budget Scenarios:

The “Received Budget Scenario” is approximately US\$ 1,054 million per annum. This allocation is optimised in the analysis to find the optimal investment split between preventive maintenance, rehabilitation and upgrading to paved standards. The maintenance activities investigated are those with identifiable intervention levels¹ and reset values², associated with the pavement performance prediction models.

The “Requested Budget Scenario” of US\$ 1,243 million was also investigated. The “Technical Needs Scenario” was investigated to determine the immediate need for preventative maintenance, rehabilitation and upgrading to paved standards.

Output:

The optimal allocation of the Received and Requested budgets with regards to the distribution of the funding towards regravelling, reseals, rehabilitation and upgrade to paved standards.

The consequences of the Received Budget on the future road network condition, asset value and vehicle operating costs.

The consequences of the Requested Budget on the future road network condition, asset value and vehicle operating costs.

The current Technical Needs of the road networks; it is the intervention treatments that are required “now”, irrespective of budget. This engineering needs determination provides a base for comparison against planned investment levels.

¹ The level at which it is necessary to intervene with a treatment, such as reseal, rehabilitation, regravell, etc.

² The value of a pavement performance measure (e.g. distress), after a treatment.

The needs analysis entails the prediction of the future performance of the road network using the current condition and pavement performance prediction models. In the analysis, realistic pavement preventive maintenance, upgrading and rehabilitation repair actions (treatments) are considered for each road analysis segment over a long-term analysis period. The premise is on keeping the roads infrastructure preventively maintained, a world-wide accepted sound infrastructure asset management principle. The effect of each repair alternative is calculated to determine the improvement in road condition and other economic consequences. The preservation of the road network is also taken into account to ensure roads with low traffic are maintained. Furthermore, the result of the analysis is multi-year preventive maintenance and rehabilitation/upgrading plans for different budget scenarios. The main items influencing the needs analysis are:

- Current network condition for paved roads (see Paragraph **Error! Reference source not found.**);
- Current network condition for unpaved roads (see Paragraph **Error! Reference source not found.**);
- Performance Models – HDM-4 (see Paragraph 7.3);
- Preventive Maintenance and Rehabilitation/ Upgrading Treatments and their Costs (see Paragraph 7.4);
- Optimisation Function (see Paragraph 7.5);
- Analysis Scenarios (see Paragraph 7.7).

The road network with condition data described in Chapter **Error! Reference source not found.** was analysed for the purposes of this report and all references to “road network” refers to the road lengths discussed there in.

7.1 Current Network Condition for paved roads

Visual assessments and roughness measurements were carried out during 2018 to record the extent of defects and performance on paved roads. These assessments

were used to determine the current condition of paved roads to serve as the starting point for the future prediction of conditions.

See Chapter **Error! Reference source not found.** for a brief discussion on the paved road network condition.

7.2 Current Network Condition for unpaved roads

Visual assessments were carried out during 2018 to record the extent of defects and performance on unpaved roads. Visual assessments determined the current quantity and quality of gravel material on the unpaved road network. The need for maintenance and regravelling is based on the predicted riding quality that can be achieved due to the characteristics and quality of the wearing course material.

See Chapter 6.3 for a brief discussion on the condition of unpaved roads.

7.3 Pavement Performance Models

7.3.1 Paved roads

To study the long-term consequences of preventive maintenance and rehabilitation actions it is important to use reliable pavement performance prediction models. The pavement performance models of HDM-4 were customised in the dTIMS decision support software for network analysis purposes.

The HDM-4 models attempt to model the complex interaction between vehicles, the environment and the pavement structure and surface. Pavement performance is principally a function of the combined effects of traffic and weather. Traffic loads induce stresses and strains within the pavement layers. The magnitude of these responses depends on the load characteristics and the layer thickness and stiffness. Under repeated loads these stresses and strains cause fatigue in bound materials and deformation of all pavement layers. Weathering and solar radiation causes asphaltic materials to age, become brittle, and more susceptible to cracking and disintegration. Once cracking has been initiated, it increases in both intensity and severity, and eventually potholes are formed. Furthermore, open cracks allow surface water to infiltrate the pavement layers hastening the process of disintegration, reducing the shear strength of the bound materials, and thus increasing the rate of deformation

under traffic. The cumulative deformation under wheel loads manifests in wheel path rutting. This in effect increases the pavement roughness, which in turn is further increased by weather and seasonal changes. Pavement roughness, the main indicator of pavement service, is therefore the result of a chain of distress mechanisms and the combination of various modes of distresses. This complex interaction between various distress types and the environment is reflected in the pavement performance models used in HDM-4.

Five HDM-4 pavement performance models have been customised into the dTIMS software. See Table 7-1 and Figure 7-1 for these distresses and their interaction.

TABLE 7-1: DISTRESSES MODELLED BY HDM-4	
DISTRESS TYPE	DESCRIPTION
All Cracking	The area of cracking greater than 1 mm in width, as a percentage of the total carriageway area.
Wide Cracking	The area of cracking wider than 3 mm in width, as a percentage of the total carriageway area.
Ravelling	The total area ravelled as a percentage of the total carriageway.
Potholing	The total area of open potholes (minimum depth of 25 mm and diameter of 150 mm) as a percentage of the total carriageway.
Rutting	The mean and standard deviation of rut depth.
Road Roughness	Roughness in IRI (International Roughness Index)

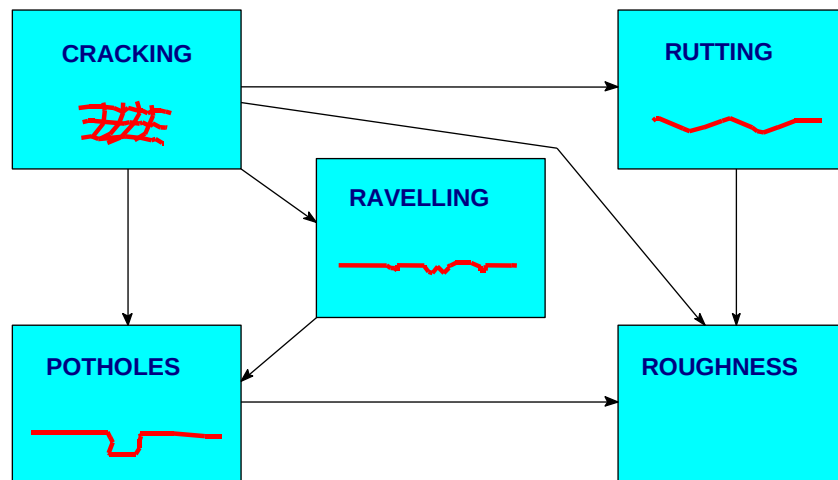


Figure 7-1: Interaction of the distress types as modelled in HDM-4

7.3.2 Unpaved roads

The needs analysis entails the prediction of the future performance of gravel roads using the current condition as the starting point in the gravel road performance models. In the analysis, realistic maintenance, regravelling and upgrading actions are considered for each road analysis segment over an analysis period. The effect of each repair alternative is calculated as well as the vehicle operating cost associated with the predicted gravel road performance. The result of the analysis is a multi-year list for routine blading maintenance, regravelling (rehabilitation of the wearing course) and upgrades to paved standards for different budget scenarios, whilst minimising the savings in costs for the road users and the road authority. The main items influencing the needs analysis are:

- present condition;
- the HDM-4 pavement performance models;
- maintenance, regravelling and upgrading treatments and their costs;
- vehicle operating cost; and
- the optimisation function.

7.4 Preventive Maintenance and Rehabilitation Treatments

In the analysis, sets of preventive maintenance and rehabilitation strategies (sequence of treatments) were considered for each road segment according to its current condition, predicted future conditions and benefits. Table describes the treatments, their triggers and costs. These unit costs are financial unit costs.

TABLE 7-2: TREATMENTS AND TRIGGERS		
TREATMENT	COST 2018 IN US\$	TRIGGER
Reseal	10 / m ²	All cracks > 2% and wide cracks ≤ 15% and roughness ≤ 4.2 IRI and rutting ≤ 20mm and AADT ≥ 4000 or Wide cracks between 5% and 10% and AADT < 4000 and roughness ≤ 4.2 IRI and rutting ≤ 20mm
Reseal Light	8 / m ²	All cracks > 1% and wide cracks ≤ 5% and roughness ≤ 4.25 IRI and rutting ≤ 20mm and AADT < 4000
Light Rehabilitation	45 / m ²	Wide cracks > 10% and condition > 45% and roughness ≤ 6.6 IRI and rutting < 20mm or Wide cracks ≤ 10% and roughness > 5.5 but < 6.6 IRI and rutting < 20mm or Roughness ≤ 5.5 and wide cracks > 10% and rutting < 20mm
Heavy Rehabilitation	60 / m ²	Roughness > 5.5 IRI and wide cracks > 10% or Roughness > 6.6 IRI or Rutting > 20mm or Overall pavement condition very poor

Regravel	13,500 / km	The option to regravel the road is triggered when the wearing course thickness is less than 30mm.
Upgrade to standard paved road	1,050,000 / km	Traffic > 200 AADT and economic merit, i.e. savings in road user costs exceed the cost of the upgrade. Total transport costs (TTC) must be positive (refer 7.5.2).

7.5 Optimisation Function

In this analysis, the maintenance plan was derived with the overall objective of maximising the condition of the road network through the Area-Under-the-condition-Curve (AUC) approach. The best possible condition under the budget constraint is thus aimed for. For unpaved roads, life cycle Total Transportation Costs (TTC) were calculated to identify feasible links for upgrading to paved standards. This ensures feasible recommendations concerning proposed upgrades to paved standards.

7.5.1 Area-under-the-condition-curve (AUC)

The area-under-the-curve benefit is calculated by summing the present value of the difference between the condition index resulting from the repair strategy (combination of repair actions over the analysis period) and the condition index for the Do-Nothing alternative for each year in the Analysis Period. The condition index is a combination of the pavement performance models of HDM-4, namely cracking, potholes, roughness, rut depth and gravel thickness. The area-under-the-condition curve benefit calculations are weighted by traffic (AADT). Figure 7-2 illustrates the method of calculating this benefit.

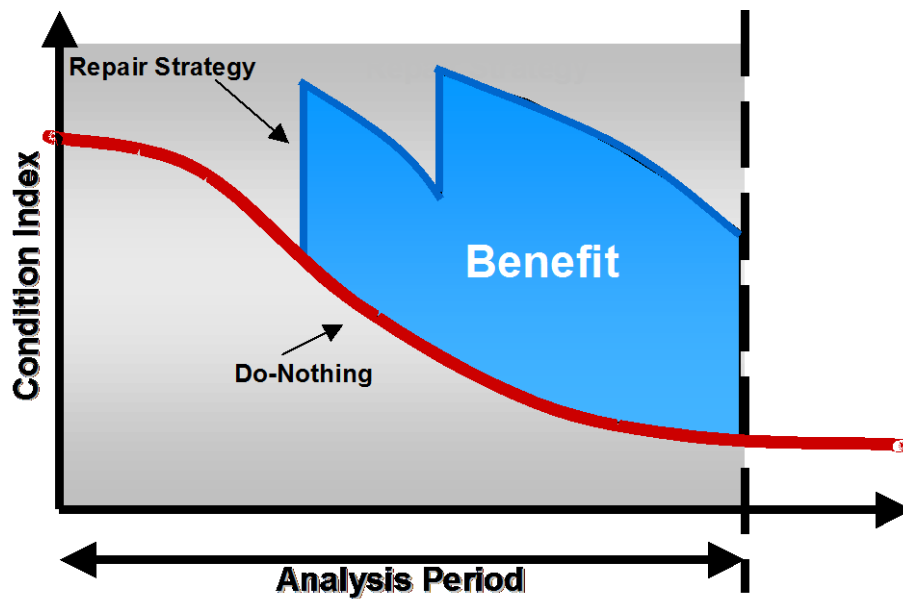


Figure 7-2: Area-Under-the-Condition Curve

The following equation represents the benefit calculation in year "i" for the area-under-the-condition curve approach.

$$\text{Benefit}(i) = \text{AADT}(i) * (\text{Str_Cond}(i) - \text{DN_Cond}(i))$$

where: i = The year in the Analysis Period

$\text{Benefit}(i)$ = Benefit of the Strategy for year i

$\text{AADT}(i)$ = AADT on the Element in year i

$\text{Str_Cond}(i)$ = Value for the repair strategy's condition index in year i

$\text{DN_Cond}(i)$ = Value for the do-nothing alternative's condition index in year i

For each strategy a Condition Index (CIND) is calculated representing the condition of the road segment over time. The area underneath the curve is the "distance" between the condition index of the Do Nothing strategy and the Repair strategy, totalled for all analysis years and weighted by traffic. Any strategy that improves the condition of the pavement will thus have a positive area underneath the curve because it is an improvement in comparison to the Do Nothing Strategy. Roads carrying high traffic receive a higher benefit than those with lower traffic. This method is used on networks where there are also roads with traffic volumes that are low.

7.5.2 Total transport cost (TTC)

Another method of optimisation is the life-cycle total transportation costs (TTC) that includes both the costs of maintaining a road network and the costs of operating vehicles upon it. Increased expenditure by the roads authority will lead to reduced cost for road users but there is theoretically a level of expenditure by the road authority at which the total transport costs are optimised. This concept is graphically illustrated in Figure 7-3. As the maintenance level (and cost) increases, the vehicle operating costs decrease. However, the increased spending on maintenance is not necessarily proportional to the savings in user costs resulting in an optimal maintenance scenario where the total transport cost to society can be minimised. Ideally a road authority should maintain a road system at the maintenance level associated with the minimum TTC.

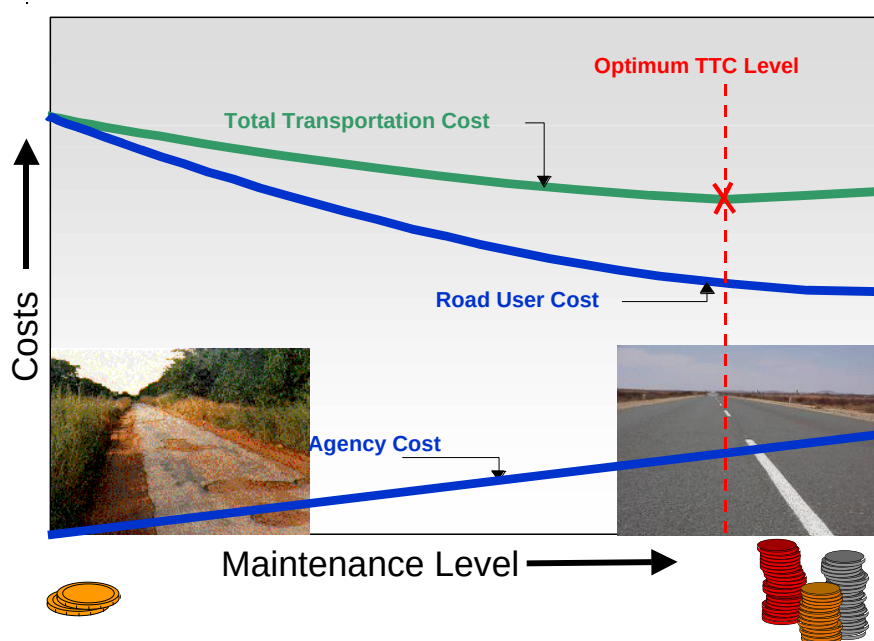


Figure 7-3: Illustration of Optimum Total Transport Costs

If this method is used for optimisation, maintenance strategies are selected to ensure the combined sum of the costs to the roads authority and road users are minimised. The savings in road user costs should always exceed the cost of the maintenance or repair strategy otherwise no economic merit exists to perform such a maintenance strategy. This minimisation of TTC therefore produces the lowest economic cost. *However, low traffic roads do not always have sufficient economic justification for*

maintenance due to limited savings in road user costs. This optimisation method is not recommended because roads without economic merit for maintenance will deteriorate beyond repair, and accessibility will be compromised. It is for this reason that the road network is optimised using the AUC method, but the TTC functions were added to determine which unpaved roads have economic merit for upgrading to paved standards.

7.6 Committed Projects

These are projects that are already earmarked for upgrading or maintenance and whose funding has already been secured and works programmed. They are spread through the analysis period in accordance with their planned programmes of executions and budgets.

Appendix A contains some of the committed projects that were included in the analysis.

7.7 Budget scenarios

The results of the analysis are highly dependent on the funding available for the maintenance of the network and all results are thus expressed as the consequence of a funding / budget scenario.

The funding scenarios discussed and analysed in this report concern the budget values for the following treatments only:

- Preventive maintenance (light reseal/reseal) of paved roads.
- Rehabilitation of paved roads (light / heavy).
- Regravelling of unpaved roads.
- Upgrading of unpaved roads to paved standards.

Other budget items, such as routine maintenance, capacity and geometric improvements, bridge maintenance and repairs, salaries, overheads, etc., are not analysed in this report and these requirements should be added to any budget needs that are calculated during this analysis.

The total budget of UNRA is US\$ 1,036 million for 2019. Another, but higher constrained budget scenario and a Do Nothing scenario were analysed to investigate the consequences of various funding levels. A Technical Needs scenario is included to determine the immediate need (backlog). All budgets investigated are discussed in Table 7-3.

TABLE 7-3: FUNDING SCENARIOS INVESTIGATED

FUNDING SCENARIO		DESCRIPTION	FUNDING LEVEL
Constrained funding scenarios	“Received Budget” US\$ 1,036 million	This scenario investigates the future network performance if an annual investment of US\$ 1,054 million is made for reseals, rehabilitation, regravelling and upgrading to paved standards.	US\$ 1,054 million per annum
	“Requested Budget” US\$ 1,243 million	This scenario investigates the future network performance if an annual investment of US\$ 1,243 million is made for reseals, rehabilitation, regravelling and upgrading to paved standards.	US\$ 1,243 million per annum
	“Do Nothing” Budget	This scenario was analysed to determine the condition and deterioration of the network if no funding was available. No committed projects were analysed as part of this scenario.	US\$ 0 million per annum
“Technical Needs Budget”		This scenario selects the first and earliest treatment that is triggered for each road segment, according to the intervention measures adopted for UNRA. This scenario is used to determine the funding level that is currently required to remove the backlog with regards to preventive maintenance, rehabilitation and upgrading to paved standards.	Funding need for the year 2019 is determined by the immediate technical needs of the road network.

7.8 Other analysis inputs

A traffic baseline growth rate of 6% was used for traffic growth on all roads in the network. This was considered the best estimate currently, given a lack of country wide growth rate data.

The analysis and planning period of the analysis was taken as 10 years.

A discount rate of 8% was used in the analysis.

Summary:

This scenario determined the funding level for UNRA roads to perform all the preventive maintenance, rehabilitation and upgrading to paved standard measures, needed from an engineering point of view, within the first analysis year.

The immediate technical needs for roads in the UNRA network are:

- ***US\$ 84 million is needed for light and heavy rehabilitation of 446 km paved roads.***
- ***US\$ 160 million is needed to reseal 2,751 km of paved roads to deter further deterioration (this includes light reseal).***
- ***US\$ 8,310 million is required for upgrading of 8,811km of unpaved roads to paved standards.***
- ***US\$ 32 million is needed for regravelling 2,355 km of unpaved roads.***

The Technical Needs scenario determines the required budget in 2019 (Year one) in order to reinstate the road network to a good condition. All backlog is removed immediately. The analysis determines which roads require immediate intervention of light reseal, reseal, rehabilitation, upgrade to paved standards or regravelling, and the analysis subsequently calculates the associated costs. The costs are then expressed as the Technical Needs Budget and it gives us an estimate of the immediate backlog with regards to the maintenance/rehabilitation/upgrading activities investigated in this report.

This scenario will have the following consequences:

- The backlog of paved roads in a poor to very poor condition will be removed,
- all unpaved roads with justifiable economic benefits will be upgraded to paved standards, and
- all unpaved roads with insufficient gravel material will be regravelled.

The optimisation analysis selected the most cost-effective alternatives for 2019 according to the most recent condition surveys that were completed on the network:

- The immediate need for paved roads is US\$ 243 million
 - US\$ 84 million is needed for light and heavy rehabilitation: This is 446 km of road where the roughness and/or structural cracking have developed beyond preventive maintenance; structural repairs are needed to improve the functional levels.
 - US\$ 160 million is needed for reseals: Approximately 2,751 km of paved roads should be resealed as a preventative measure to deter further deterioration.
- The immediate need for unpaved roads is US\$ 8,342 million (± US\$ 8,3billion)
 - US\$ 8,310 million is required for upgrading of 8,811 km of unpaved roads to paved standards.
 - US\$ 32 million is needed for regraveling: In total 2,355 km of unpaved roads require replacement of the gravel wearing material. The road user costs on gravel roads with no wearing course material are normally excessive.

Figure8-1 shows the funding level required to remove the backlog and Figure 8-2 shows the length of road affected by this funding scenario.

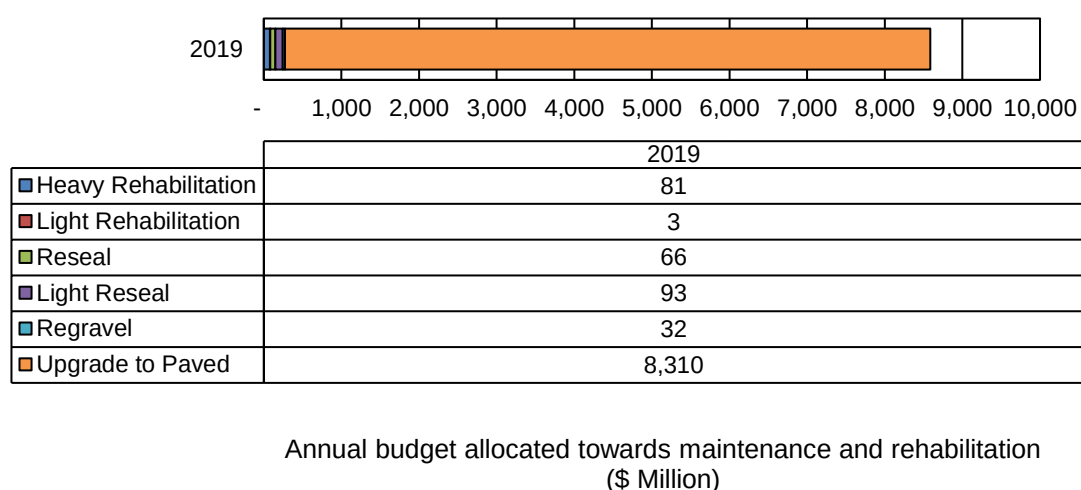


Figure8-1: Immediate Technical Needs 2019 (cost, US\$ million)

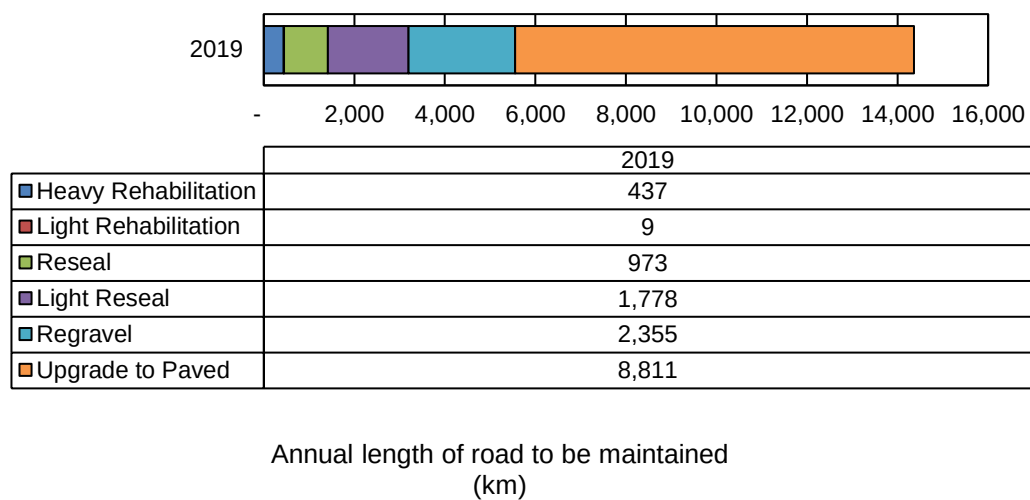


Figure 8-2: Immediate Technical Needs 2019 (length, km)

Summary:

This scenario determined the optimised funding allocation of the received budget towards preventative maintenance, rehabilitation and upgrading to paved standards. This is a constrained funding scenario with US\$ 1,037 million available of which US\$ 534 million is allocated to committed projects in the first year of the analysis.

The following results are relevant

- *US\$ 76 million is available annually for light and heavy rehabilitation of paved roads. Over a five year period it is possible to rehabilitate 1,231km.*
- *US\$ 26 million is available annually for reseals. 2,296 km can be resealed in five years.*
- *US\$ 928 million is available annually for upgrading of unpaved roads to paved standards. Over a five year period it is possible to upgrade 5,296 km.*
- *US\$ 25 million is available annually for regravelling. 9,204 km can be regravelled over a five year period.*

The Received Funding scenario determines the optimum allocation of the funds available. This is a constrained funding scenario with US\$ 1,054 million per annum of which US\$ 534 million is allocated to committed projects in the first year of analysis. Therefore, the optimisation of only US\$ 503 million is considered for the first year. Table 9-1 shows the fund allocations towards committed projects for paved and unpaved roads as well as the funds available for optimisation from 2019 up to 2023.

TABLE 9-1: COMMITTED FUNDS AND FUND ALLOCATIONS AVAILABLE FOR OPTIMISATION

YEAR	COMMITTED PAVED (US\$ MILLION)	COMMITTED UNPAVED (US\$ MILLION)	FUNDING AVAILABLE FOR OPTIMISATION
2019	81	453	503
2020	77	292	739
2021	31	186	872
2022	31	13	972
2023	31	-	989

Figure 9-1 shows the cost allocation towards different maintenance and rehabilitation strategies from 2019 to 2023 and Figure 9-2 shows the length of road affected by the different strategies. It should be noted that the committed projects are included in these figures.

The optimisation analysis selected the most cost-effective alternatives according to the most recent condition surveys that were completed on the network:

- The optimisation analysis determined that on average 10% (US\$ 102 million) should be allocated on an annual basis towards paved roads over a five year period.
 - On average US\$ 76 million of the US\$ 102 million should be invested towards the annual rehabilitation of roads and the remaining US\$ 26 million should be used for resealing.
 - In total over the five-year period 1,231 km of road could be rehabilitated and 2,296 km could be resealed.
- On average 90% (US\$ 953 million) of the received budget should be allocated annually to unpaved roads.
 - US\$ 928 million should be used for the yearly upgrading of unpaved roads to paved standard and the remaining US\$ 25 million should go towards regravelling.
 - In total over the five year period 5,296 km of road could be upgraded to paved road standards and 9,204 km could be regravelled.

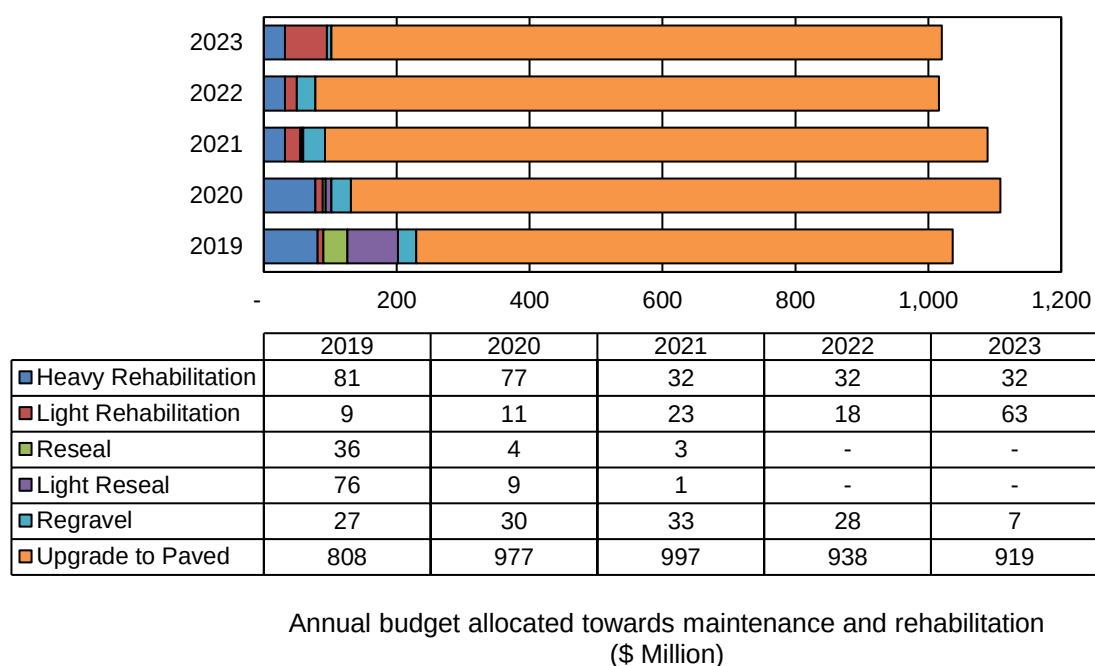


Figure 9-1: Received Funding Scenario optimised allocation (cost, US\$ million)

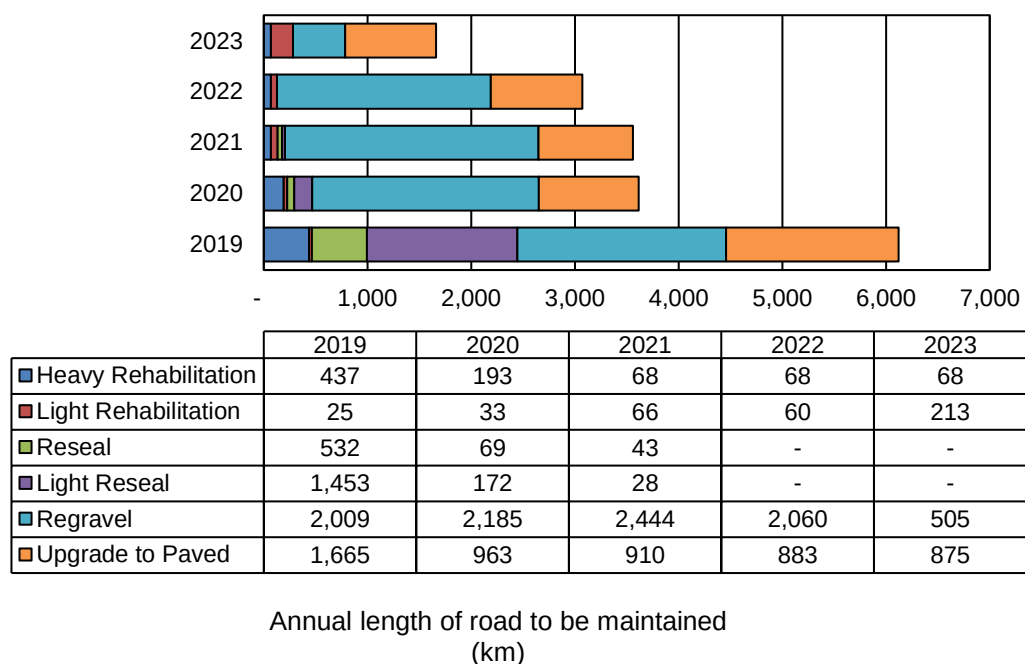


Figure 9-2: Received Funding Scenario optimised allocation (length, km)

Summary:

This scenario determined the optimised funding allocation of the requested budget towards preventative maintenance, rehabilitation and upgrading. This is a constrained funding scenario with US\$ 1,243 million available per annum of which US\$ 534 million is allocated to committed projects in the first year of the analysis.

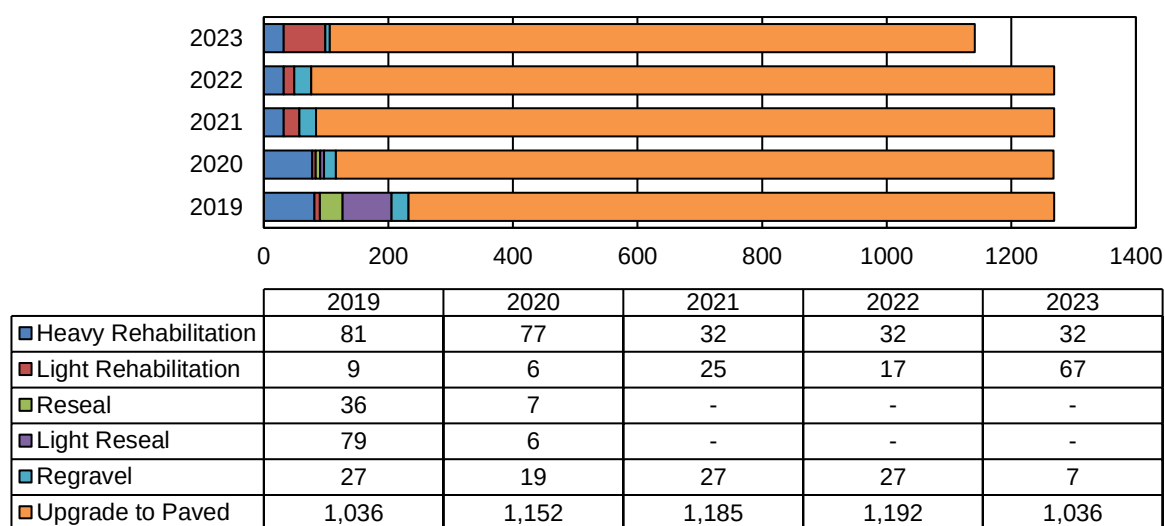
The following results are relevant

- *US\$ 76 million is available annually for light and heavy rehabilitation of paved roads. Over a five year period it is possible to rehabilitate 1,232 km.*
- *US\$ 26 million is available annually for reseals. 2,273 km can be resealed in five years.*
- *US\$ 1,120 million is available annually for upgrading of unpaved roads to paved standards. Over a five year period it is possible to upgrade 6,213 km.*
- *US\$ 21 million is available annually for regravelling. 7,944 km can be regravelled over a five year period.*

The requested funding scenario of US\$1,243 million was analysed and the cost optimisation is shown in Figure 10-1. Figure 10-2 shows the length of road that can be maintained or rehabilitated with the requested funding. It should be noted that the committed projects were included in this analysis. Table 10-1 summarises the fund allocations towards committed projects for paved and unpaved roads as well as the funds available for optimisation from 2019 up to 2023.

TABLE 10-1: COMMITTED FUNDS AND FUND ALLOCATIONS AVAILABLE FOR OPTIMISATION

YEAR	COMMITTED PAVED (US\$ MILLION)	COMMITTED UNPAVED (US\$ MILLION)	FUNDING AVAILABLE FOR OPTIMISATION
2019	81	453	735
2020	77	292	899
2021	31	186	1,052
2022	31	13	1,225
2023	31	-	1,110



Annual budget allocated towards maintenance and rehabilitation
(\$ Million)

Figure 10-1: Requested Funding Scenario optimised allocation (cost, US\$ million)

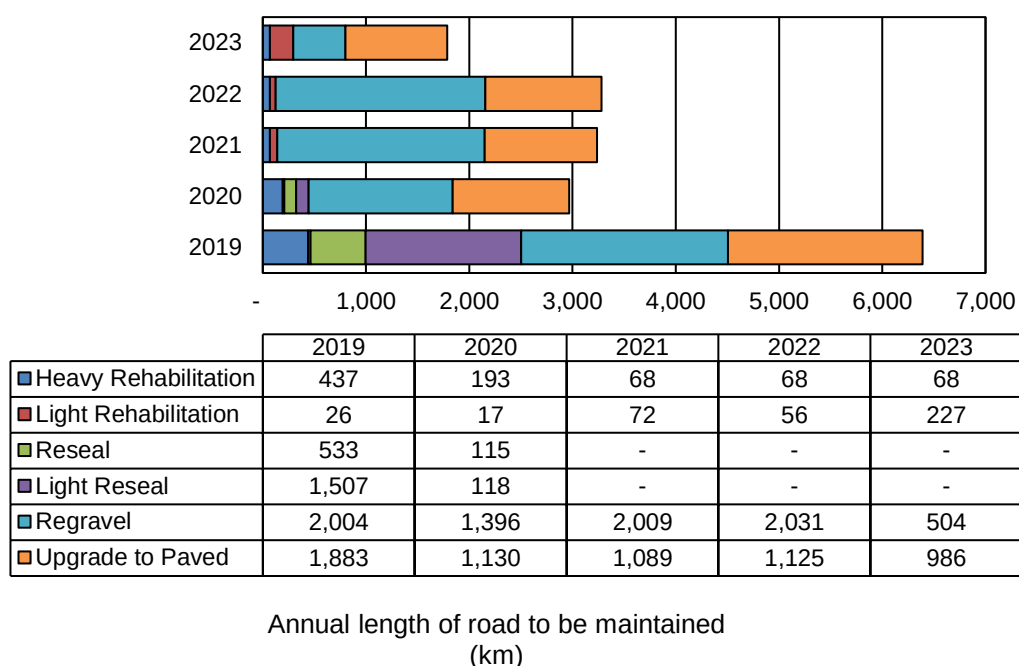


Figure 10-2: Requested Funding Scenario optimised allocation (length, km)

The optimisation analysis selected the most cost-effective alternatives according to the most recent condition surveys that were completed on the network:

- The optimisation analysis determined that on average 8% (US\$ 101 million) should be allocated on an annual basis towards paved roads over a five year period.
 - On average US\$ 76 million of the US\$ 101 million should be invested towards the annual rehabilitation of roads and the remaining US\$ 25 million should be used for resealing.
 - In total over the five year period 1,232 km of road could be rehabilitated and 2,273 km could be resealed.
- On average 92% (US\$ 1,142 million) of the Requested Budget should be allocated to unpaved roads annually.
 - US\$ 1,120 million should be used for the yearly upgrading of unpaved roads to paved standards and the remaining US\$ 21 million should be allocated towards regraveling.
 - In total over the five year period 6,213 km of road could be upgraded and 7,944 km could be regravelled.

11 Comparison of Different Funding Levels

Summary:

- *The Received Funding Scenario is sufficient to maintain and improve the condition of the paved road network.*
- *The optimum fund allocation towards paved roads is very similar for the Received and Requested Funding scenarios.*
- *Budget allocations of higher funding levels would favour the maintenance and upgrading of unpaved roads.*
- *For a funding level below US\$ 534 million the distribution of funds is towards committed projects which mainly account for rehabilitation of paved roads and the upgrading of unpaved roads.*
- *Over a period of 10 years the average gravel thickness will remain around 130 mm for both the Received and Requested Funding Scenarios.*
- *On average the Received Budget will regravell 14% more length of road than the Requested Budget and the Requested Budget will upgrade 15% more road length than the Received Budget.*

11.1 Split between Paved and Unpaved roads

Figure 11-1 shows the allocation towards Paved and Unpaved roads for the three funding levels. It is clear from the graphs that all 3 budget scenarios allocated close to the same amount to paved roads. Therefore, as the budgets increase the funding allocation towards unpaved roads increase.

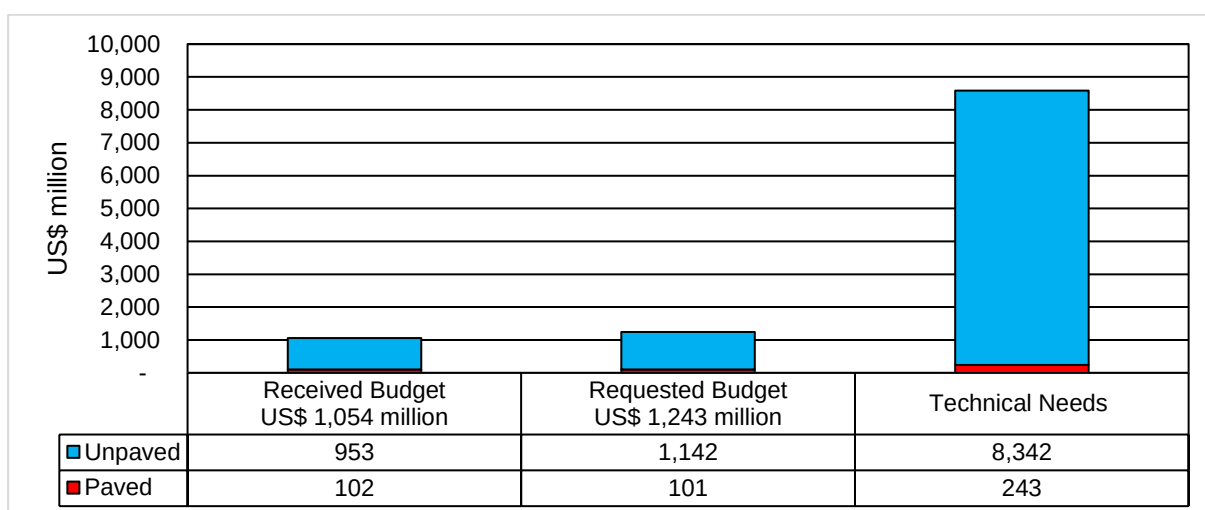


Figure 11-1: Funding split between paved and unpaved roads for three budget scenarios

11.2 Predicted Condition of Paved Roads

During the optimisation analysis, a predicted condition index (CIND) is calculated using the performance models for rut depth, roughness, cracking and potholes. This index has a rating scale of 0% (very poor) to 100% (very good) and is used to provide management with decision support with regards to the future predicted condition under the different funding scenarios. The predicted network condition in Figure 11-2 is the average predicted condition for the paved road network, weighted by the length of the roads.

The predicted condition for the Received, Requested and Technical Needs Scenarios are similar over the next ten years. The predicted current condition of the road network is fair and the Received Funding level is sufficient to maintain and even improve the network condition. Therefore any funding scenario such as the Requested and Technical Needs scenarios with higher funding levels will direct additional funds towards the improvement or upgrading of unpaved roads. This can be seen in Figure 11-1 where the funding allocated towards paved roads remains relatively constant for all funding scenarios and the allocation towards unpaved roads increases as more funds become available.

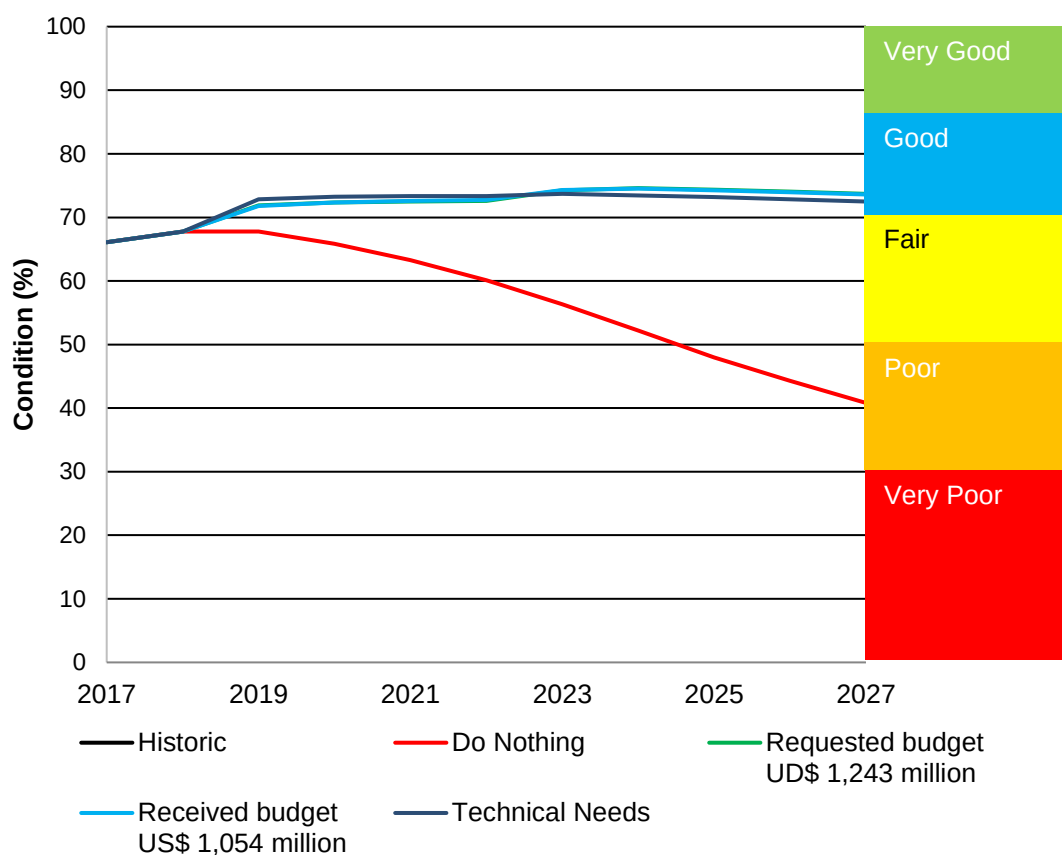


Figure 11-2: Predicted condition for paved roads

Figure 11-3 shows the distribution of the predicted network condition into five condition categories. From an engineering point of view; roads in a poor to very poor condition require expensive repair measures such as rehabilitation to restore the functionality of the roads. Ideally this proportion should be kept at a level not more than ten percent of the total road length, which is also advocated by World Bank.

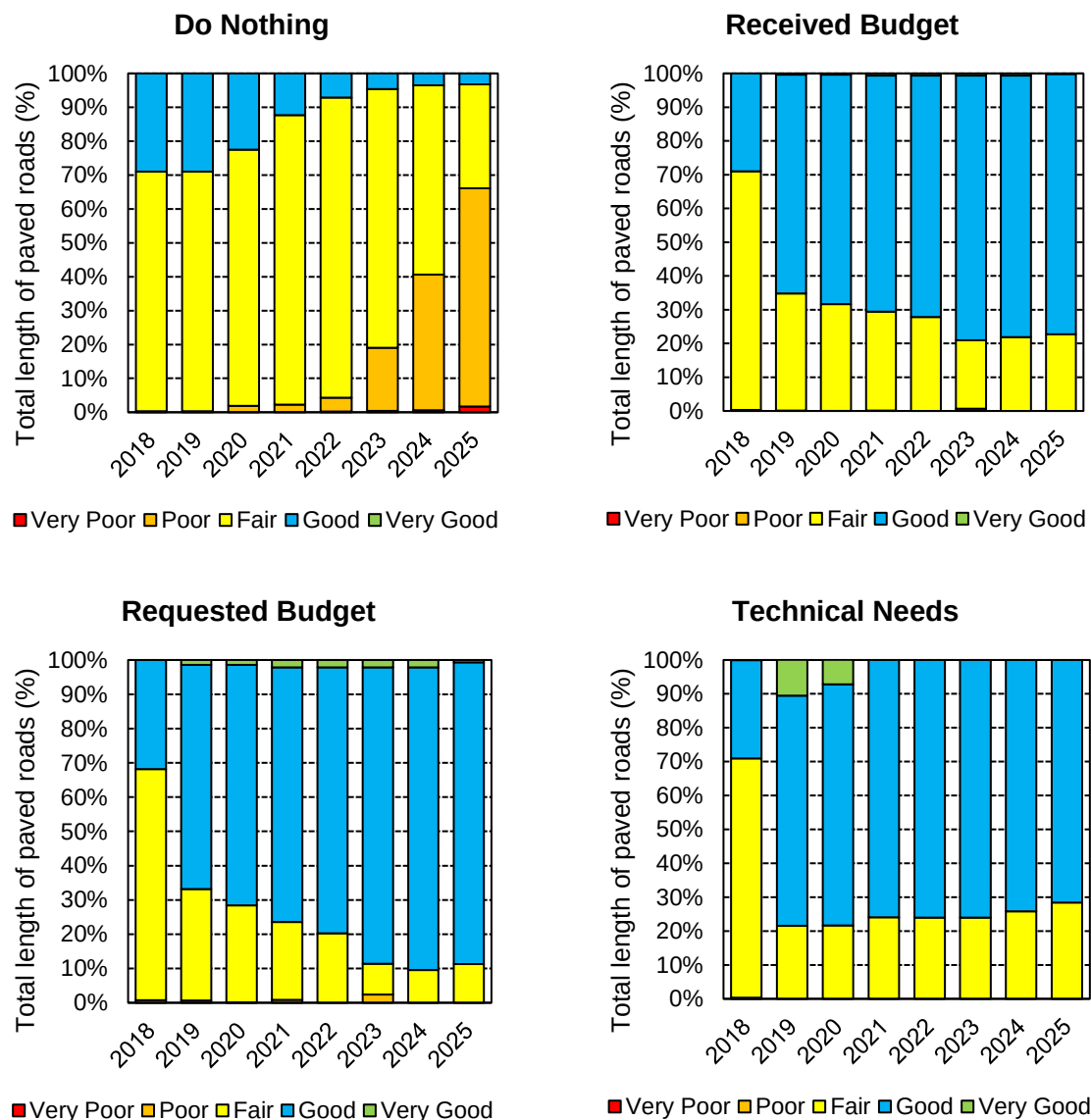


Figure 11-3: Predicted condition distribution for paved roads

11.3 Predicted Gravel Thickness of Unpaved Roads

Gravel thickness is used to measure the health of the unpaved road network. The roughness model of unpaved roads is dependent on the quality and quantity of the gravel wearing course material which influence the performance of the unpaved roads. During the optimisation process, regravelling is recommended as a preventive measure to ensure unpaved roads are passable and road user costs are kept at acceptable levels. Under constrained funding scenarios the optimisation analysis favours paved roads initially to ensure these roads are maintained. Once the condition

of paved roads is at acceptable levels, more funding is directed towards unpaved roads.

All three funding scenarios follow the same trend. Over a period of 10 years the average gravel thickness is around 130 mm. As the funding level increases the allocation between regravelling and upgrading leans toward the upgrading of unpaved roads to paved standards.

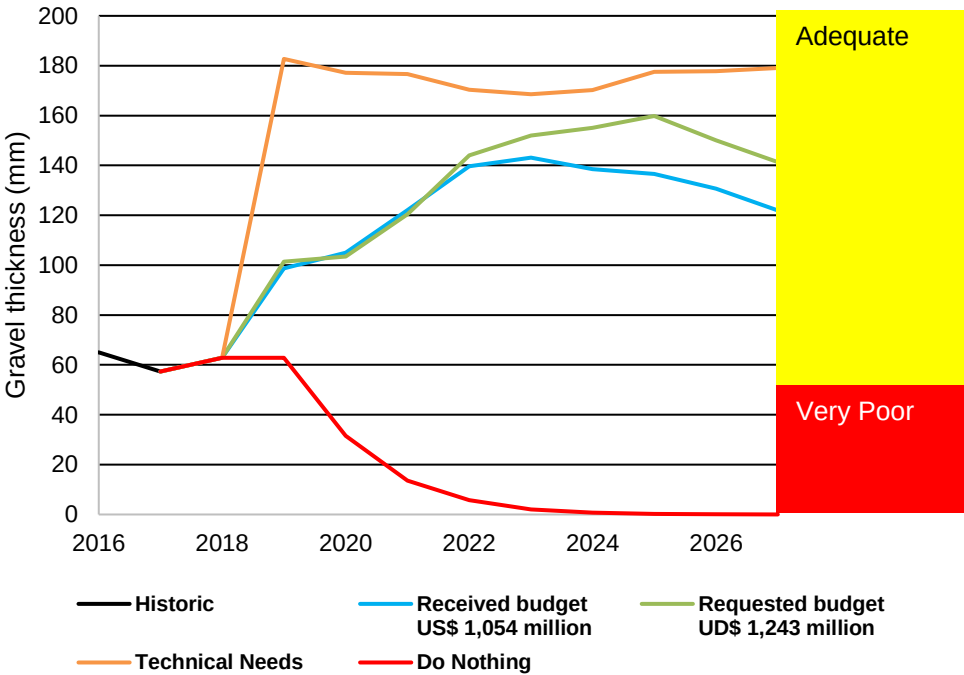


Figure 11- 4: Predicted gravel thickness on unpaved roads

Figure 11-5 shows the distribution of gravel thickness for the funding scenarios. As seen with the different funding scenarios, roads with inadequate wearing course material decrease as funding increases. A reason for this phenomenon is that as funding increases, more roads get upgraded to paved standards as well as money being available to gravel the remaining unpaved roads.

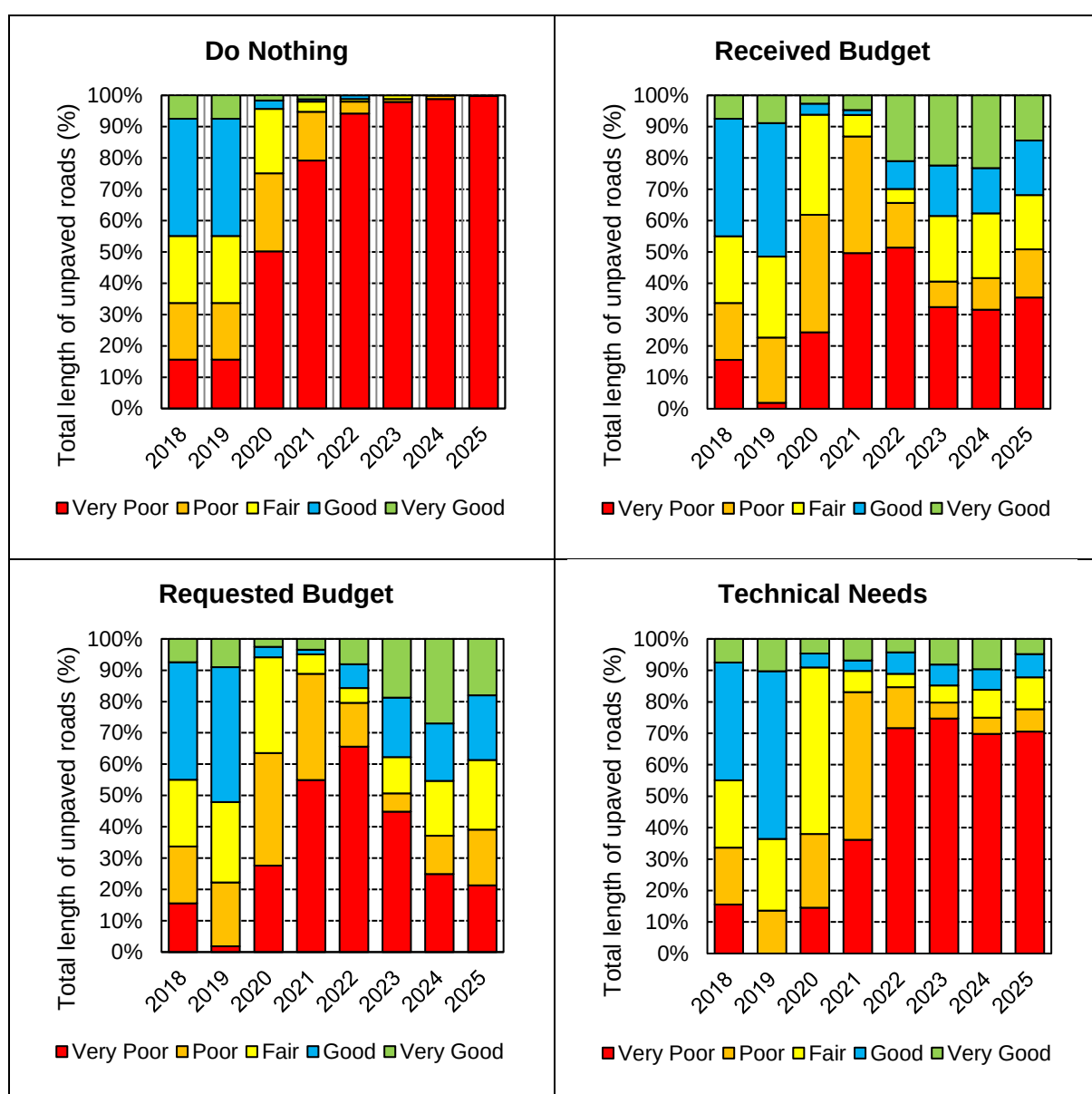


Figure 11-5: Distribution of predicted gravel road condition

11.4 Cost and Length Comparison of Received and Requested Funding

Since the predicted condition of the network for the Received and Requested Funding Scenarios is almost similar for the analysis period, the cost and length distributions for paved and unpaved roads were assessed in detail. Figure 11-6 shows the funding level for paved roads and the affected lengths. For the Received Funding Scenario an average of US\$ 102 million annually is used to treat 3,528 km over a five year period. This is very similar to the US\$ 101 million used on the Requested Funding Scenario to treat 3,506 km of paved road in five years.

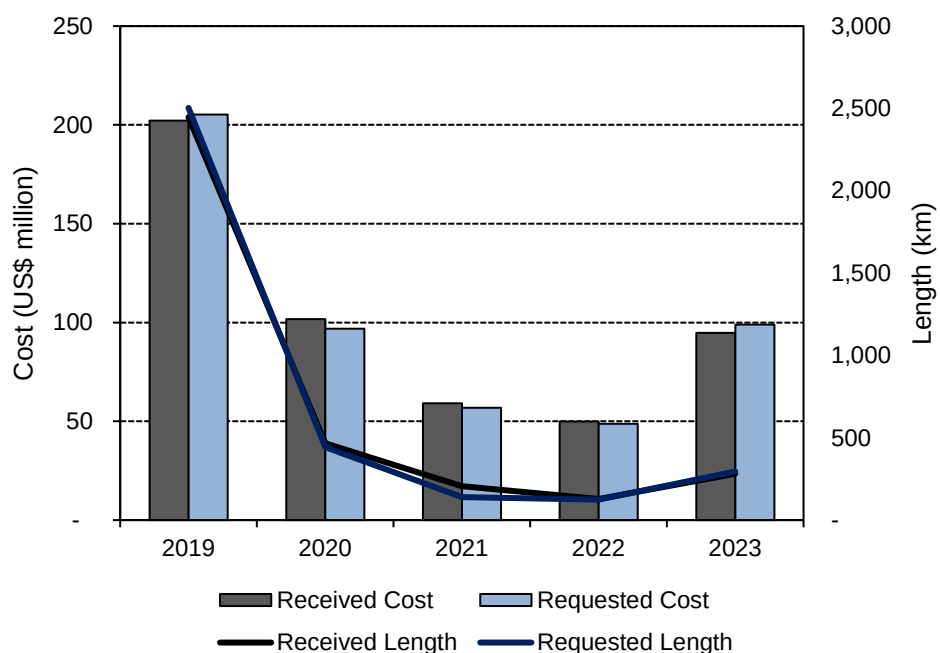


Figure 11-6: Comparison of Received and Requested Funding Scenarios and the associated length of the paved network affected

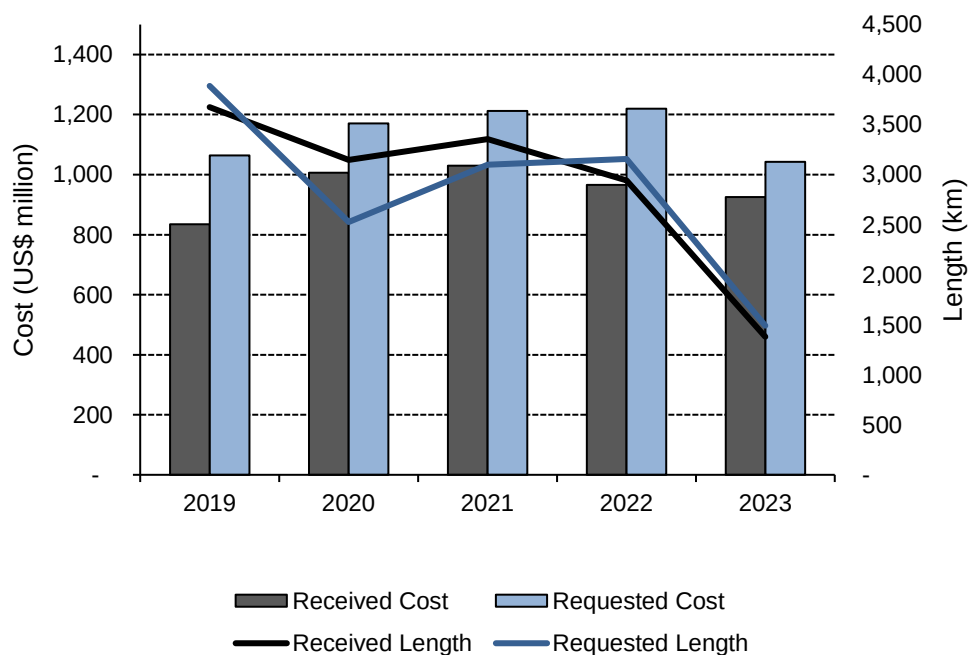


Figure 11-7: Comparison of Received and Requested Funding Scenarios and the associated length of the unpaved network affected

Figure 11-7 shows the funding levels for unpaved roads and the affected lengths. For the Received Funding Scenario an average of US\$ 953 million is used annually to treat

14,500 km over a five year period. The Requested Funding Scenario uses US\$ 1,142 million to treat 14,157 km over a five year period. The split of funds between regravelling and upgrading was assessed and is shown in Figure 11-8 and Figure 11-9.

Figure 11-8 indicates that the Received Funding Scenario allocates more funds towards regravelling than the Requested Funding Scenario. US\$ 25 million is annually allocated towards the regravelling of 9,204 km of unpaved roads over a period of five years on the Received Budget. The Requested Budget allocates US\$ 21 million annually towards the regravelling of 7,944 km of road.

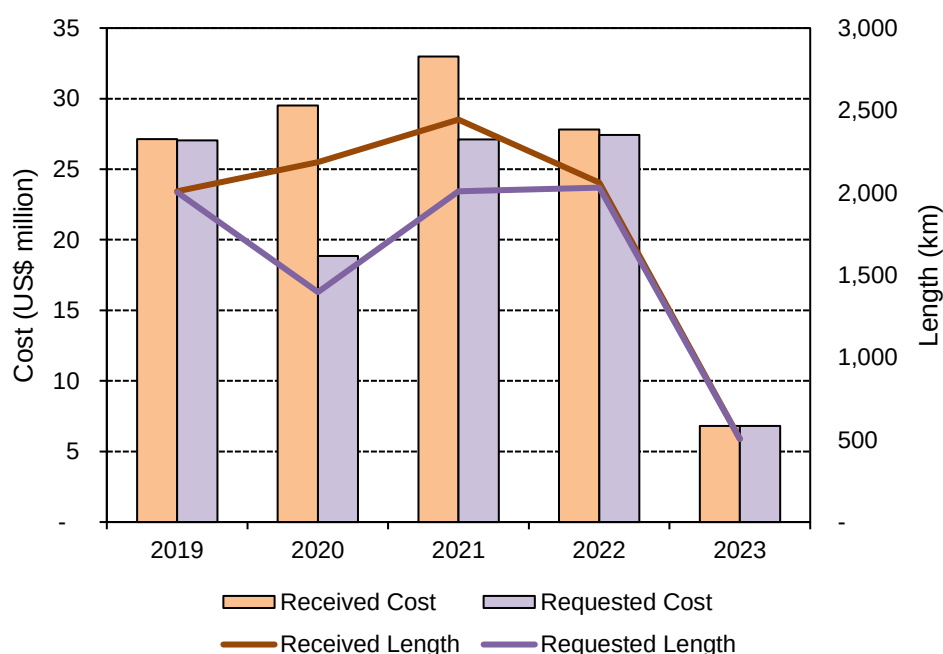


Figure 11-8: Comparison of Received and Requested Funding Scenarios and the associated length of the unpaved network regravelled

Figure 11-9 shows that the Requested Budget allocates more funds towards the upgrading of unpaved roads than the Received Budget. The Received Budget allocates US\$ 928 million annually for the upgrading of 5,296 km over a period of five years whereas the Requested Budget allocates US\$ 1,120 million annually towards the upgrading of 6,213 km over a five year period.

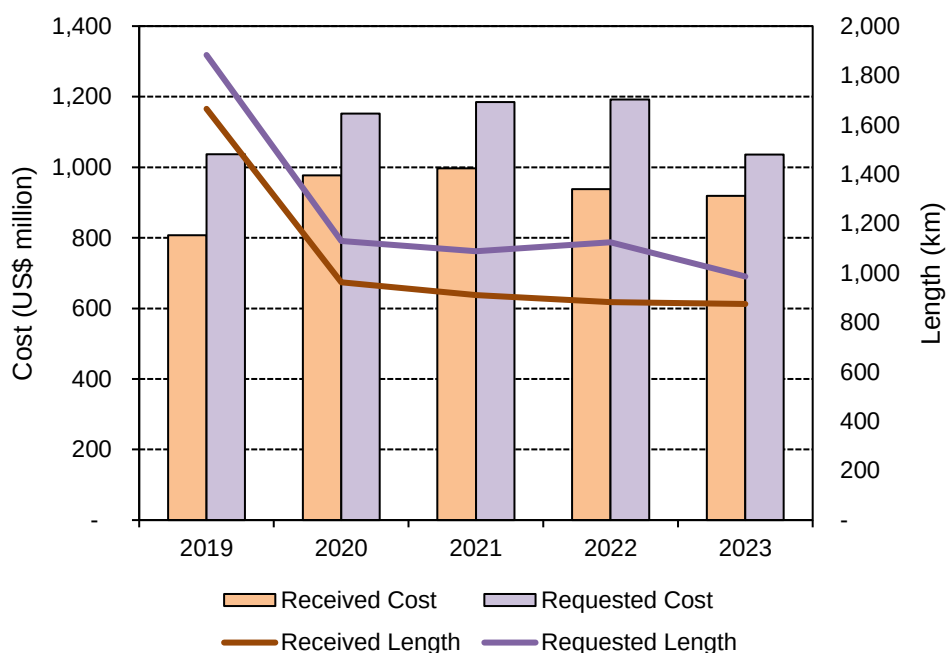


Figure 11-9: Comparison of Received and Requested Funding Scenarios and the associated length of the unpaved network upgraded

The current condition of paved roads is good therefore fund allocations lean more towards the upgrading of heavily trafficked unpaved roads. Therefore if higher funding levels are available a larger portion of the fund allocation would go towards upgrading of unpaved roads.

Summary:

The current VOC is US\$ 1,276 per 1000 vehicle- kilometres travelled.

VOC by 2027 for:

- **Do Nothing scenario is US\$ 1,286**
- **Received Budget is US\$ 802**
- **Requested Budget is US\$ 751**

The current average VOC per thousand vehicle-kilometres driven on the UNRA road network is calculated at approximately US\$ 1,276 (or US\$1.276 per vehicle-kilometre). If nothing is done to the road network the average VOC will increase to US\$ 1,286 per 1000 vehicle-kilometres by 2027. This value could be reduced to US\$ 802 per 1000 vehicle-kilometres travelled by 2027 should the Received Budget be adopted. This value further decreases to US\$ 751 if the Requested Budget is allocated. Figure 12-1 shows the trend for VOC on the road network.

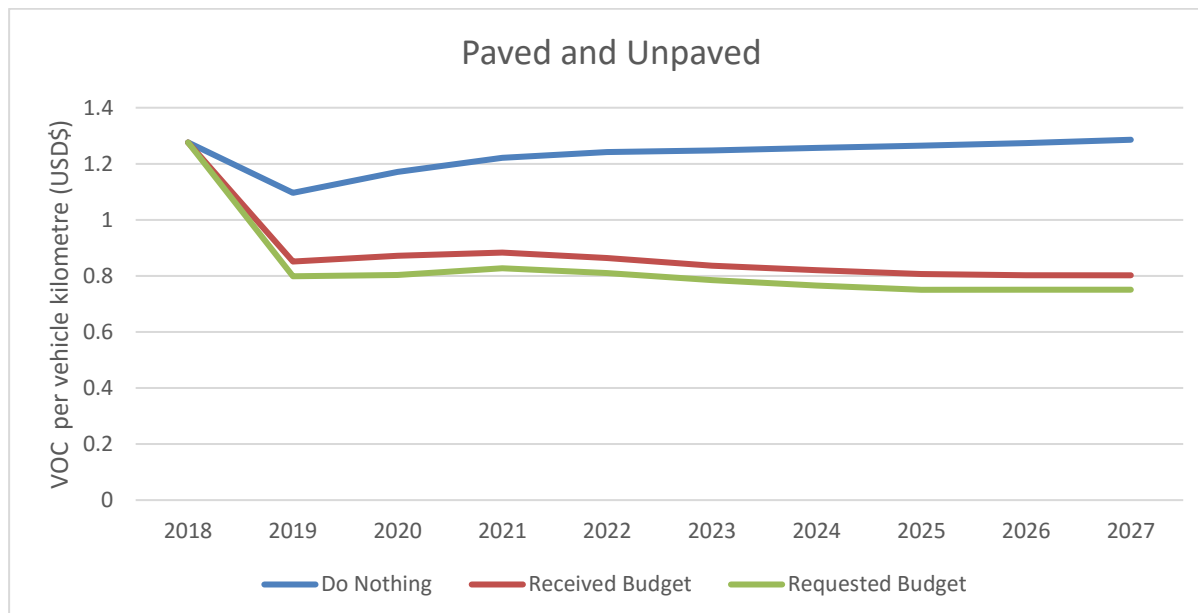


Figure 12-1: Vehicle operating cost per 1000 vehicle-kilometres travelled on the UNRA road network

Summary:

The current EUC is US\$ 1,101 per 1000 vehicle-kilometres travelled.

The EUC by 2027 for:

- **Do Nothing scenario is US\$ 1,709**
- **Received Budget is US\$ 331**
- **Requested Budget is US\$ 327**

Excess User Costs are defined as the excess or additional user costs that would not have been experienced if the road had acceptable roughness. Excess user cost is calculated on roads with an average IRI of greater than 3.1 IRI. The current Excess User Cost per thousand vehicle-kilometres driven on the UNRA road network is calculated at approximately US\$ 1,101. If nothing is done to the road network the average EUC will increase to US\$ 1,709 per 1000 vehicle-kilometres by 2027. This value could be reduced to US\$ 331 and US\$ 327 per 1000 vehicle-kilometres travelled by 2027 for the Received Budget and Requested Budget respectively. Figure 13-1 shows the trend for EUC on the road network.

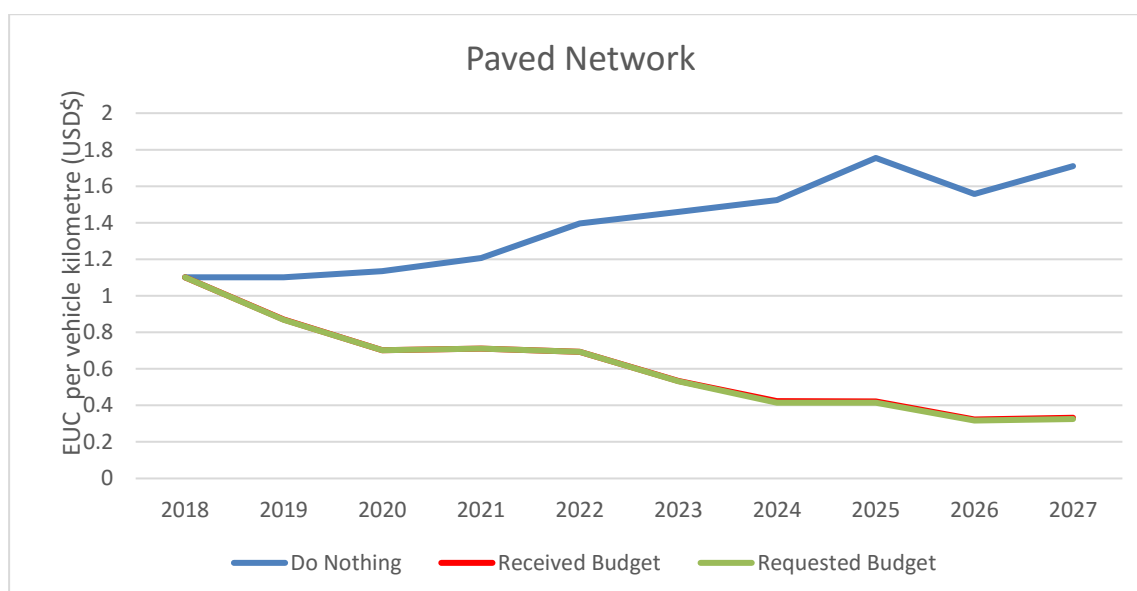


Figure 13-1: Excess user cost per 1000 vehicle-kilometres travelled on the UNRA road network

Summary:

The Current Depreciated Replacement Cost (CDRC) for the paved road network is US\$ 3,2 billion.

The Current Replacement Cost (CRC) for the UNRA paved network is US\$ 4,7 billion.

For the unpaved network the CDRC is US\$ 649 million and the CRC is US\$ 1,549 million

Asset value is a measure of the worth of a road network and the value of a pavement consists of the following costs:

- The ground preparation work,
- the road foundations and
- the road structural layers.

Over time, pavement structures deteriorate and the value of the structural layers decreases accordingly. Eventually, reconstruction (i.e. heavy rehabilitation or regravelling) becomes necessary and if the deterioration of the pavement layers continues even further a stage will be reached where the pavement has no functionality and the value of the structural layers is zero.

The value of any road is calculated as the value it would have if it was new, minus the cost of taking it from its present condition to the “very good” or new condition. The amount subtracted represents the cost of eliminating any deficiency the road may have. This concept is similar to the generally accepted accounting practice of calculating an asset’s “book value”, which equates to cost minus accumulated depreciation. Asset value is expressed as the Current Depreciated Replacement Cost (CDRC), i.e. in terms of current US\$ values. The potential or maximum asset value refers to the replacement cost in 2018 US\$ values, or Current Replacement Cost (CRC).

Note that the cost of road furniture such as signs, bridges and culverts are not included in this calculation of asset value.

It is estimated that the Current Replacement Cost, CRC of the paved road network is approximately US\$ 4,667 million (US\$ 4,7 billion); thus assuming all roads are in a very good condition. The asset value associated with the current condition of the paved road network, i.e. the Current Depreciated Replacement Cost (CDRC), was calculated at US\$ 3,206 million (US\$ 3,2 billion). The CDRC is currently 69 per cent of the CRC, or alternatively stated there is asset loss of 31%. A CDRC to CRC ratio greater than 80% is indicative of a well-maintained road network and the current performance of the paved road network is therefore fine. Figure shows the predicted CRC and CDRC for the paved portion of the UNRA network.

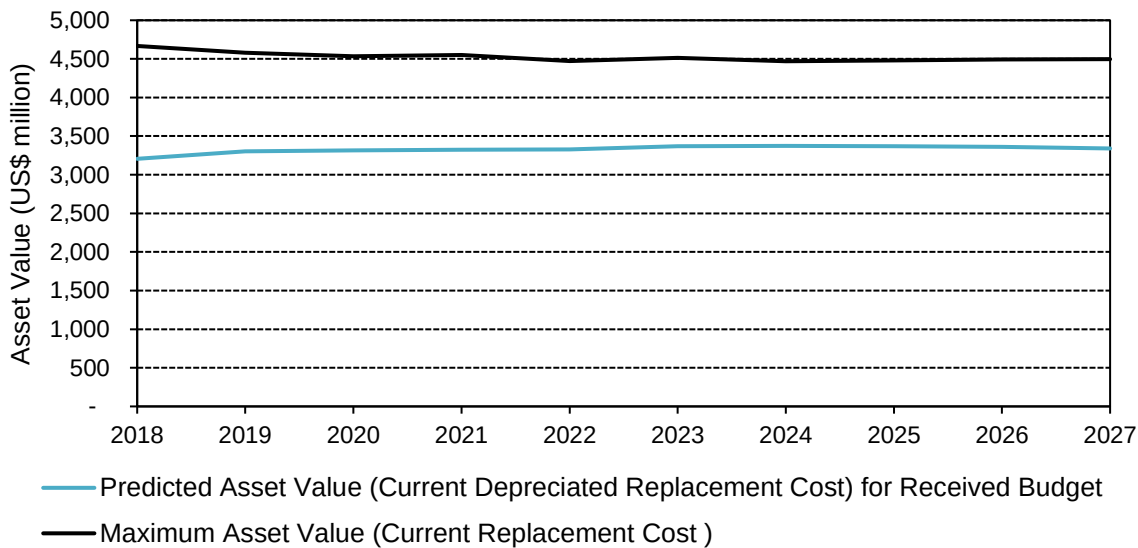


Figure 14-1: Predicted Asset Value for the Received Budget versus the Maximum Asset Value for Paved Roads

It is estimated that the Current Replacement Cost, CRC of the unpaved road network is approximately US\$ 1,550 million; thus assuming all gravel roads have plenty of gravel material. The asset value associated with the current gravel thickness of the unpaved road network, i.e. the Current Depreciated Replacement Cost (CDRC), is approximately US\$ 649 million. The CDRC is currently 42 per cent of the CRC.

It can be noted that the Maximum Asset value and the CDRC of the unpaved roads increase over the analysis period. This is due to current unpaved roads that are upgraded to paved standards during the analysis.

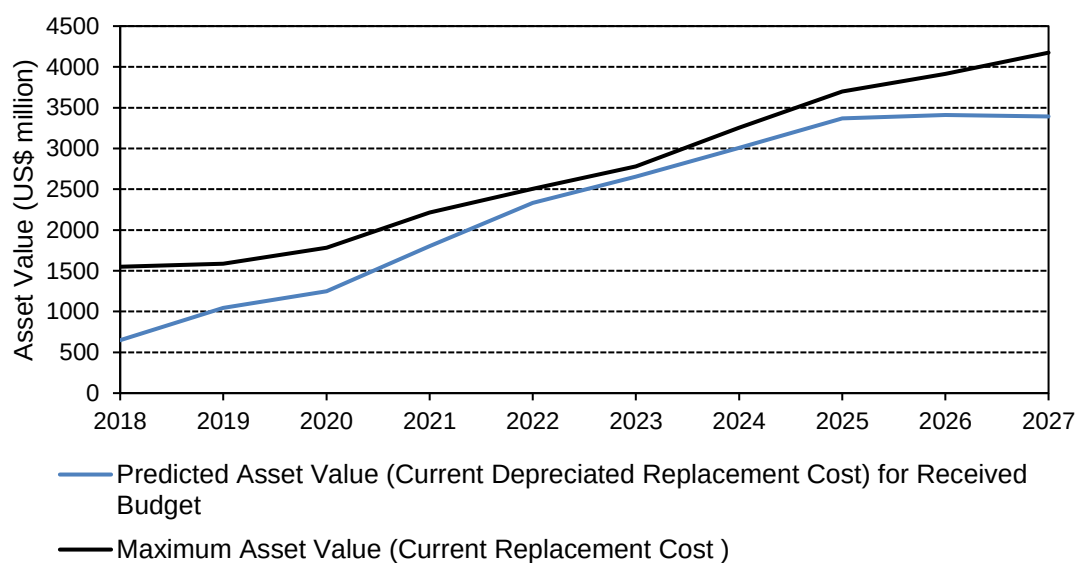


Figure 14-2: Predicted Asset Value for the Received Budget versus the Maximum Asset Value for Unpaved Roads

15 Summary

This report concludes with a summary of the most important findings of the strategic analysis.

This analysis has highlighted interesting aspects of the UNRA road network. These are summarised in bullet points below for each of the paved and unpaved road networks.

Paved roads

- The paved road network comprises 4,683 km, which is only 23% of the total UNRA road network.
- 84% of the paved network was visually assessed and was analysed in this strategic analysis.
- The analysed network carries over 50% of all traffic, and 7,742 million vehicle-kilometres are travelled annually on this network.
- 11% of the network carries more than 10,000 vehicles per day, and 68% carries less than 5,000 vehicles per day.
- The overall average VCI of the paved network is 76%, which indicates a network in “good” condition. Compared to the previous analysis, there has been an increase of 1% in the overall condition.
- The percentage roads in poor and very poor condition (the backlog) is 7%, which is less than the World Bank recommended maximum of 10% for this category. The backlog for rehabilitation is therefore insignificant.
- The percentage roads in fair condition is 22%, which is 3% less than the experienced based recommended maximum of 25%. This shows that good preventive maintenance strategies are being followed.
- A Technical Needs analysis of reseal, light and heavy rehabilitation needs, given current road conditions and covering 5 years from 2019, identified needs to the value of US\$ 416 million on 3,595 km of road. This is only US\$ 83 million per annum.

- Under the Received Funding Scenario, the optimisation allocated on average US\$ 102 million per annum to paved roads over a five year period. This represents 10% of the total available funding of this funding scenario, allowing 1,231 km road to be rehabilitated, and 2,296 km road to be resealed over the five years.
- The result of the allocation of the Received Funding Scenario is that the current paved road condition will remain “good” over the 5 years. Excess User Costs will be reduced to US\$ 534 per 1000 vehicle-kilometres which is a 50% decrease achieved for road users compared to the current EUC.
- Maintenance of the full paved network should be strived for given its important function. This is indeed achievable under the Received Funding Scenario. The optimisation analysis indicated that under higher funding scenarios, such as the Requested Funding Scenario, the optimum allocation to paved roads is almost similar to the optimum allocation of the Received Funding Scenario.

Unpaved roads

- The unpaved network of UNRA is extensive and amounts to 16,111 km of road. This represents 77% of the total network.
- 89% of the unpaved network was visually assessed and was analysed in this strategic analysis.
- The analysed network carries 46% of all traffic, and 6,561 million vehicle-kilometres are travelled annually on this unpaved network.
- 69% of the network carries more than 500 vehicles per day. 41% of unpaved road travel concerns volumes between 1,000 and 5,000 vehicles per day. This implies a high need for upgrading to paved standards.
- The average VCI of the unpaved network is 61%, which indicates a “fair” condition. Compared to the previous analysis, the condition has largely remained the same.
- The percentage roads in poor and very poor condition is 25% which is 15% higher than the World Bank recommended maximum of 10% for this category. The backlog for regravelling is therefore significant.
- The average gravel thickness of the unpaved network is 62 mm.

- A Technical Needs analysis of investment requirements for regravelling and upgrading to paved standards, given current road conditions and traffic volumes, and covering 5 years from 2019, identified investment requirements to the value of US\$ 9,845 million on 12,579 km of road.
- The regravelling need, based on the Technical Needs determination, is US\$ 32 million, or 2,355 km approximately over the 5 year period.
- The upgrading from unpaved to paved road investment need, based on the Technical Needs determination, is US\$ 9,813 million, or 10,224 km approximately over the 5 year period.
- Under the Received Funding Scenario, the optimisation allocated on average US\$ 953 million per annum to unpaved roads over a five year period. This represents 90% of the total available funding of this funding scenario, allowing 5,296 km road to be upgraded to paved standards, and 9,204 km road to be regravelled over the five years.
- The optimisation analysis indicated that under higher funding scenarios than the Received Funding Scenario, such as the Requested Funding Scenario, the optimum allocation to paved roads is similar to the optimum allocation of the Received Funding Scenario, but more funds are available for the high need for upgrading of unpaved roads to paved standards.

Table 15-1 compares the optimum funding allocations of Received Budget with the Requested Budget scenario.

Table 15-1: Funding allocation of the Requested Budget Compared to Received Budget (US\$ million and km summed for the first 5 years)		
	Received Budget	Requested Budget
Rehabilitation allocation	US\$ 378 (1,231 km) US\$ 38 per annum	US\$ 378 (1,232 km) US\$ 38 per annum
Reseal allocation	US\$ 130 (2,296 km) US\$ 13 per annum	US\$ 129 (2,273 km) US\$ 13 per annum
Regravelling allocation	US\$ 124 (9,204 km) US\$ 25 per annum	US\$ 107 (7,944 km) US\$ 21 per annum

Upgrade to paved roads allocation	US\$ 4,639 (5,296 km) US\$ 928 per annum	US\$ 5,601 (6,213 km) US\$ 1120 per annum
TOTAL PER ANNUM	US\$ 1,054	US\$ 1,243

Assumptions made

1. The unit cost per km includes all costs to construct a road
2. There will be no excessive delayed payments and thus no unplanned interest paid to service providers
3. Inflation will be fairly stable as predicted
4. Traffic growth is uniform all through at 6%
5. The discount rate is 8%
6. Supporting processes are efficient
7. All construction processes are undertaken in one year for the delivery of those triggered lengths
8. Traffic assignment is accurate
9. That land is available in time

Appendix A – Committed Projects

Appendix B – Maintenance, Upgrading and Rehabilitation Plan (M&R's